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Fractional Topological Phases in Two-Dimensional Moiré Systems

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Master's Thesis

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Abstract

Moiré materials, in particular transition metal dichalcogenide homobilayers, offer a versatile, tunable platform to study strongly correlated phases, including topological fractional quantum Hall-like states. This thesis studies a general model of moiré material of Kwan et al. [23], with the two valleys modelled through two Landau levels of opposite Chern number, coupled via a tunable inter-valley coupling α, and with a single-particle dispersion of bandwidth W. We extend the model to include projection onto higher Landau levels, and an inverted single-particle dispersion. It is shown analytically and numerically, that the resulting phase diagrams are symmetric under this dispersion–inversion. Using exact diagonalization on a torus, we map out the phase diagram at fillings $\nu = 1/2$, $2/3$, 1 in the lowest Landau level as a function of the model's parameters and of the screening parameters of the Coulomb interaction. At $\nu = 2/3$, through ground-state degeneracy and spectral flow, we identify a valley-polarized fractional quantum Hall state (3-fold degenerate) and a valley-unpolarized Laughlin-like fractional topological insulator (9-fold degenerate), both of which survive finite bandwidth (up to $|W| \approx 0.1$) and non-ideal inter-valley coupling (down to $\alpha_c \approx 0.72$). At $\nu = 1$ and $\nu = 1/2$, the expected integer quantum Hall and composite-Fermi-liquid behavior is recovered, along with plausible signatures of an inter-valley coherent state. At all fillings, Fermi liquid state is confirmed at large single-particle dispersion W. An analytic Hartree–Fock approximation of the valley-depolarization transition is developed and found to reproduce the qualitative trends of the numerical phase boundaries, albeit with a systematic overestimate of the critical parameters by up to an order of magnitude. Extending the calculation to the first excited Landau level at $\nu_{n=1} = 1$, using tilted boundary conditions to mitigate finite-size anisotropy, we find evidence for a non-abelian fractional topological insulator composed of two time-reversed Moore–Read states (with 4-fold and 36-fold ground-state degeneracy for odd and even particle number, respectively), whose stability against bandwidth, inter-valley coupling, and screening is examined, recovering comparable trends and comparable stability against bandwidth, but also significantly larger inter-valley coupling stability (down to $\alpha_c \approx 0.64$, and $\alpha_c \approx 0.12$ at finite Rytova–Keldysh screening) than for the abelian lowest Landau level fractional topological insulator. For both the abelian and non-abelian fractional topological insulator states, the Rytova–Keldysh screening length l_{RK} is found to have a non-monotonic effect on stability: a maximum of α-stability is reached at finite l_{RK}, at the cost of significantly reduced stability against bandwidth W. These results establish a systematic numerical account of how bandwidth, valley coupling, and screening govern the competition between fractional and integer quantum Hall states, valley polarization, and other competing phases in moiré TMDs, including non-abelian states.

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Chapter 1

Introduction

1.1 Moiré materials

Moiré materials are two-dimensional heterostructures formed by stacking atomically thin layers with a small lattice mismatch or twist angle between them. Due to this misalignment, a spatial interference pattern is formed between the two lattices, giving rise to an artificial crystal with a substantially larger lattice constant — the moiré lattice. In the small-angle approximation, the moiré lattice constant is $a_m \approx a_0/\theta$, so for typical twist angles $\theta \in (0, 4^\circ)$ the lattice constant increases by up to two orders of magnitude relative to the underlying lattice constant a_0. This in turn leads to effective folding of the Brillouin zone (BZ) of the original material, and emergence of moiré bands in the band structure. Due to the significant separation of length scales between the moiré lattice and the underlying atomic lattice, the effect of interactions is strongly enhanced relative to the kinetic energy, giving rise to a wide range of strongly correlated phases of matter.

Additionally, properties of these materials can be very widely tuned through:

- A wide choice of constituent 2D materials — graphene, transition metal dichalcogenides (TMDs), transition metal oxides [29] — and their combination into multilayer structures
- The twist angle between layers, which sets the moiré lattice constant a_m, the interlayer hybridization, and hence the band structure
- Strain engineering [16]
- Screening of Coulomb interactions via dielectric environment engineering [58]
- Continuous *in situ* control of the electron filling of each moiré unit cell

The last two points are easiest to understand in terms of a typical device geometry (Fig. 1.1), where the material is sandwiched between two spacers and two conducting gates [5]. The dielectric properties of the spacers and the distance of the gates from the sample directly affect the screening of electron–electron interactions within the active layer. Independently, because the moiré unit cell is so much larger than the atomic one, tuning the gate potential allows the electron filling of each moiré unit cell to be continuously controlled, enabling doping on the order of 10 electrons per unit cell.

This combination of enhanced interactions and wide tunability of properties has made moiré materials one of the central topics in modern condensed matter physics.

1.1.1 Transition metal dichalcogenides

While the field of moiré materials was catalyzed by the discovery of superconductivity and correlated insulating behavior in twisted bilayer graphene (TBG) [6, 7], semiconducting materials joined the field shortly after [30]. As noted above, the large moiré lattice constant and

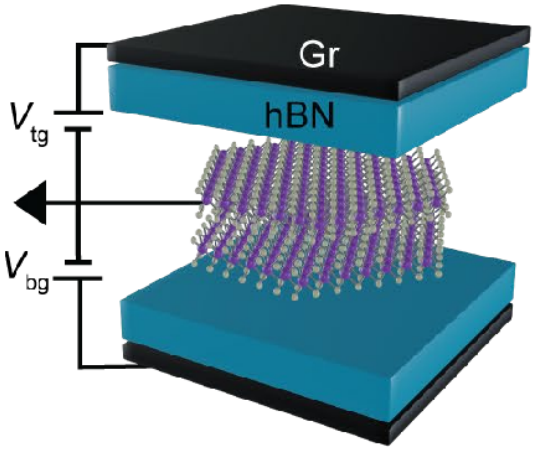
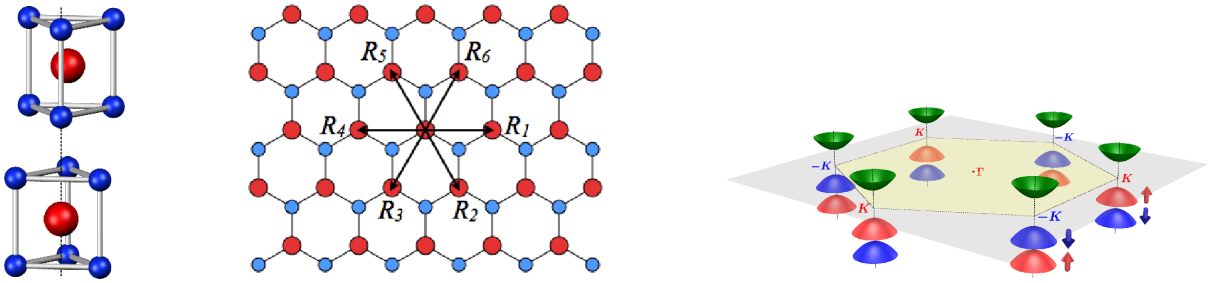


Figure 1.1: Typical experimental setup for a moiré device: the active material is sandwiched between two dielectric spacers and two conducting gates. Adapted from [5].

the associated BZ folding effectively confine electrons and suppress their kinetic energy. The resulting flat bands are a defining feature of moiré materials. In graphene-based heterostructures, however — described by massless Dirac fermions with no energy gap [35] — flat bands emerge only at special "magic" angles. In semiconductors, the nonzero bandgap relaxes this condition [53]: magic angles can still be identified [14], but the material remains gapped over a wide, continuously tunable range of twist angles. Within this class, homobilayers of TMDs are of particular interest because of their

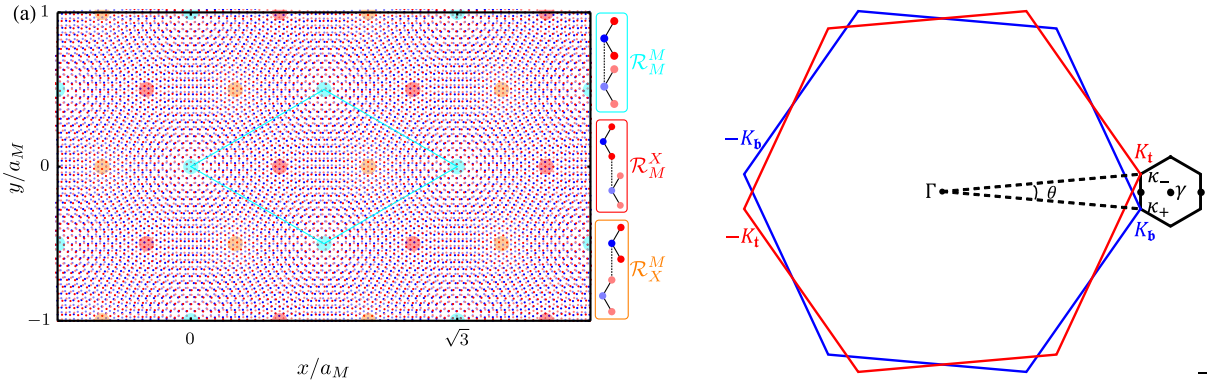
- Relatively simple structure (two layers of the same material)
- Relatively low disorder density
- Direct bandgap
- Spin-valley locking [30].



(a) (b) (c)

Figure 1.2: Structure of a transition metal dichalcogenide monolayer. (a) Atomic coordination: the transition metal (red) sits in trigonal-prismatic coordination with the chalcogen atoms (blue). (b) Top-view real-space structure of a TMD monolayer in the 2H phase. (c) Valence-band maxima and conduction-band minima near the K and K' points of the Brillouin zone; arrows indicate electron spin. Adapted from Ref. [54].

TMD monolayers consist of three atomic planes: a central plane of transition-metal atoms sandwiched between two planes of chalcogen atoms. The two common structural phases are 1T and 2H, with 2H being the stable phase for most commonly studied group-VI TMDs (Fig. 1.2a).



(a) (b)

Figure 1.3: Real- and reciprocal-space structure of a twisted TMD homobilayer, adapted from [37]. (a) Real-space moiré superlattice; cyan, red, and orange dots mark the $\mathcal{R}_M^M$, $\mathcal{R}_M^X$, and $\mathcal{R}_X^M$ local stacking registries (inset), with the moiré unit cell shown in cyan. (b) Hexagonal Brillouin zones of the top (red) and bottom (blue) monolayers and of the twisted bilayer (black), for twist angle θ.

In the 2H phase the transition metal sits in trigonal-prism coordination, forming a honeycomb lattice with D_{3h} symmetry (broken inversion symmetry compared to the bulk) (Fig. 1.2b). This real-space honeycomb structure gives rise, upon twisting two layers, to the triangular moiré superlattice and folded hexagonal Brillouin zone shown in Figure 1.3. Already at the level of a single monolayer, however, the honeycomb lattice — as in graphene — gives rise to two inequivalent corners of its hexagonal Brillouin zone, conventionally labeled K and $-K$ (or K and K') and known as *valleys*. The low-energy physics can be described near these two time-reversal-conjugate points, where the valence-band maxima and conduction-band minima are located [9, 22]. Xiao et al. [54] modeled this with a two-band Hamiltonian in which electrons behave as massive Dirac fermions. The bands near the K-points are built from partially filled transition-metal d-orbitals,

$$|\phi_c\rangle = |d_{z^2}\rangle, \qquad |\phi_v^\tau\rangle = \frac{1}{\sqrt{2}}\left(|d_{x^2-y^2}\rangle + i\tau|d_{xy}\rangle\right), \tag{1.1}$$

where $\tau = \pm 1$ is the valley index and the subscripts c, v denote the conduction and valence bands, respectively. The conduction-band orbital $|\phi_c\rangle$ has z-component of orbital angular momentum $L_z = 0$ and is therefore (approximately) spin-degenerate, whereas the valence-band orbital $|\phi_v^\tau\rangle$ has $L_z = \pm 2\hbar$ and is split by strong spin-orbit coupling — a consequence of the broken inversion symmetry. This produces spin-valley locking: the low-energy valence-band physics at each valley can be described by a single spin species, with opposite spins locked to opposite valleys (Fig. 1.2c), so that the overall model remains time-reversal invariant. Due to the large separation of the valleys in momentum space and spin-valley locking, inter-valley scattering is suppressed, and charge is separately conserved within each valley. This yields an emergent $U(1) \times U(1)$ symmetry under which the valley index τ is a good quantum number.

1.2 Hall effect and topological order

1.2.1 Integer quantum Hall effect

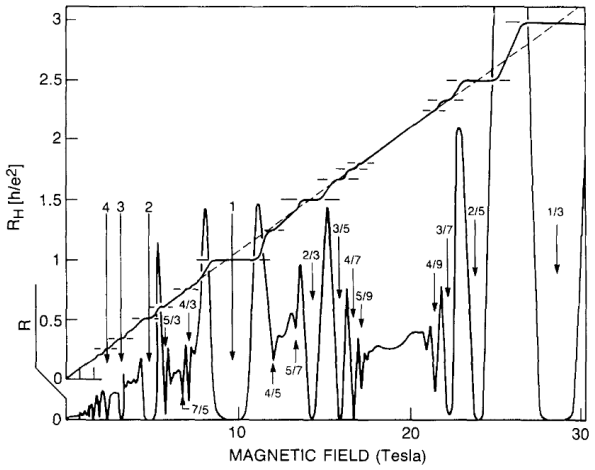


Figure 1.4: Hall resistance R_H versus magnetic field, showing the quantum Hall plateaus (labeled by filling fraction). Adapted from [47].

The physics of correlated phases in moiré materials turns out to be closely tied to that of the quantum Hall effect — the response of a two-dimensional electron gas to a perpendicular magnetic field. Classically, electrons in a 2D system subject to a magnetic field simply experience a Lorentz force, resulting in an off-diagonal Hall conductivity $\sigma_{xy} \propto B$. At low temperature and high field, however, the response becomes quantum mechanical. The magnetic field enters the single-particle Hamiltonian through the vector potential,

$$\hat{H} = \frac{1}{2m}(\hat{\mathbf{p}} - e\mathbf{A})^2, \qquad B = \nabla \times \mathbf{A}. \tag{1.2}$$

In an appropriate gauge (e.g., the Landau gauge $\mathbf{A} = (0, Bx, 0)$ or the symmetric gauge $\mathbf{A} = (-\frac{1}{2}By, \frac{1}{2}Bx, 0)$), the Schrödinger equation is solved exactly and the spectrum is found to be quantized into equally spaced, highly degenerate Landau levels (LLs). This degeneracy reflects a complete quenching of the kinetic energy — a result analogous to the flattening of moiré bands discussed above. The degeneracy of each Landau level is proportional to the number of flux quanta N_ϕ threading the system, and the filling factor is defined as $\nu = N_e/N_\phi$, where N_e is the number of electrons; varying the magnetic field therefore changes the LL filling. Linear response theory then shows that the Hall conductivity is quantized,

$$\sigma_{xy} = \nu \frac{e^2}{h}, \qquad \nu \in \mathbb{Z}, \tag{1.3}$$

producing the characteristic plateaus of Fig. 1.4 as a function of magnetic field. This effect, discovered in 1980 [50], is known as the Integer Quantum Hall Effect (IQHE).

Crucially, the IQHE is robust against local disorder and weak perturbations, because it is topological in origin: the quantized Hall conductivity in Eq. (1.3) can be related directly to a topological invariant of the system, the Chern number, as we now discuss.

1.2.2 Berry phase and Chern number

Let $\hat{H}(\mathbf{R}(t))$ be a Hamiltonian depending on a time-dependent parameter set $\mathbf{R}(t)$, with instantaneous eigenstates $|n(\mathbf{R})\rangle$ and eigenvalues $E_n(\mathbf{R})$ satisfying $\hat{H}(\mathbf{R})\,|n(\mathbf{R})\rangle = E_n(\mathbf{R})\,|n(\mathbf{R})\rangle$.

Let $|\Psi(0)\rangle = |n(\mathbf{R}(0))\rangle$ be a non-degenerate eigenstate at $t = 0$, separated from all other eigenstates by a finite energy gap. The time evolution of $|\Psi\rangle$ obeys the time-dependent Schrödinger equation,

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = \hat{H}(\mathbf{R}(t))\,|\Psi(t)\rangle\,. \tag{1.4}$$

By the adiabatic theorem, if the parameters vary slowly compared with the inverse energy gap, $|\Psi(t)\rangle$ remains an instantaneous eigenstate of $\hat{H}$ up to a phase,

$$|\Psi(t)\rangle = e^{i\gamma(t)}\,|n(\mathbf{R}(t))\rangle\,, \tag{1.5}$$

with

$$\gamma(t) = -\frac{1}{\hbar}\int_0^t E_n(\mathbf{R}(t'))\,dt' + \int_{\mathbf{R}(0)}^{\mathbf{R}(t)}\mathbf{A}(\mathbf{R})\cdot d\mathbf{R}. \tag{1.6}$$

The first term in Eq. (1.6) is the familiar dynamical phase, accumulated from the energy of the state during time evolution. The second term is the geometric Berry phase: it is independent of how fast the path in parameter space is traversed, but depends on the path itself. It is expressed as a line integral of the Berry connection,

$$\mathbf{A}(\mathbf{R}) = i\langle\Psi(\mathbf{R})|\nabla_{\mathbf{R}}|\Psi(\mathbf{R})\rangle, \tag{1.7}$$

with the associated Berry curvature $\mathbf{\Omega}(\mathbf{R}) = \nabla_{\mathbf{R}}\times\mathbf{A}(\mathbf{R})$. Crucially, when the path in parameter space is closed, the initial and final wavefunction must represent the same physical state, therefore coinciding up to a phase. Since the wavefunction is single-valued, this constrains the accumulated geometric phase, and writing it as a surface integral of the Berry curvature over the enclosed surface S gives

$$\gamma = \int_S \mathbf{\Omega}(\mathbf{R})\cdot d\mathbf{S} = 2\pi C, \tag{1.8}$$

where $C \in \mathbb{Z}$ is the Chern number. Because C is an integer, it cannot change continuously, and can change only when the energy gap to the reference state closes, violating the assumption of the adiabatic theorem — hence its topological robustness. While the Berry phase is generally gauge-dependent, both the Berry phase and the Chern number are gauge-invariant for a closed path, which is what makes C a well-defined topological invariant.

Although introduced here in general terms, this formalism directly reproduces the IQHE: taking the parameter space to be crystal momentum $\mathbf{k}$ over the (magnetic) Brillouin zone, the Berry curvature $\mathbf{\Omega}(\mathbf{k})$ plays a role analogous to a fictitious magnetic field in $\mathbf{k}$-space, the Berry connection $\mathbf{A}(\mathbf{k})$ that of a vector potential, and the Chern number that of the number of flux quanta threading the Brillouin zone. Linear response theory then shows that the Chern number determines the Hall conductivity directly,

$$\sigma_{xy} = C\,\frac{e^2}{h}, \tag{1.9}$$

so that $C = \nu$ in the IQHE of Eq. (1.3) [49].

1.2.3 Fractional quantum Hall effect

In sufficiently clean samples at strong magnetic field, additional Hall plateaus appear at *fractional* LL fillings — the Fractional Quantum Hall Effect (FQHE), directly visible in Fig. 1.4. Unlike the IQHE, the FQHE cannot be explained for non-interacting electrons: it is a direct consequence of electron–electron interactions, which lift the massive degeneracy of many-body states within a partially filled LL.

As with the IQHE, the FQHE plateaus can be related to a Chern number, provided the ground-state degeneracy characteristic of the fractional state is properly accounted for. The

most prominent plateaus occur at filling $\nu = 1/q$ with q an odd integer (Fig. 1.4). While no closed-form solution exists for the Coulomb problem, the Laughlin wavefunction [27] — the exact ground state of a simplified pseudopotential Hamiltonian — captures the essential physics and, numerically, retains remarkably high overlap with the ground state of the realistic Coulomb problem.

More generally, at filling $\nu = p/q$ with $p, q \in \mathbb{N}$, q odd, and $\gcd(p, q) = 1$, the algebra of magnetic translation operators [3] fixes the ground-state degeneracy on the torus to be q-fold, with magnetic translations mapping the q degenerate ground states into one another. These states are therefore equivalent and each carries an equal share p/q of the manifold's total Chern number, giving the fractional quantization

$$\sigma_{xy} = \frac{p}{q}\frac{e^2}{h} = \nu\frac{e^2}{h}. \tag{1.10}$$

Beyond quantized conductivity, the FQHE hosts anyonic excitations: quasiparticle and quasihole excitations carrying a fraction of the electron charge and obeying fractional exchange statistics. When two indistinguishable quasiparticles are exchanged, their wavefunction can acquire an arbitrary phase, $|\Psi\rangle \to e^{i\theta}|\Psi\rangle$. In three dimensions, the path traced by exchanging the particles twice can always be continuously deformed to a point, where by continuity the phase must be trivial, forcing $2\theta = 2\pi m$, $m \in \mathbb{Z}$ — giving a θ phase of π or a minus sign for fermions, and 0 or plus sign for bosons. In two dimensions, however, this deformation is topologically obstructed — a path encircling another particle cannot be shrunk past it — so the exchange phase can take any value $\theta \in [0, 2\pi)$. Such particles, exhibiting exchange phase distinct from fermions and bosons, are called abelian anyons. If instead the local Hilbert space of a quasiparticle is higher-dimensional, exchange corresponds to a more general unitary transformation $\hat{\mathbf{U}}$ acting on that space, $|\Psi\rangle \to \hat{\mathbf{U}}|\Psi\rangle$; since such exchanges generally do not commute, these are *non-abelian* anyons. Because the FQHE's topological character is robust to perturbations, so is the anyonic statistics of its quasiparticles, making these systems attractive as a platform for topological quantum computation — a notion of using the non-abelian anyons as qubits, where execution of quantum gates would correspond to simple spatial exchanges of the anyons — "braiding" [13, 33]. In this aspect, Laughlin states in a FQH system have abelian statistics with the exchange phase $\theta = \frac{\pi}{q}$ for $\nu = \frac{1}{q}$.

1.2.4 Composite fermions and the composite Fermi liquid

A complementary picture of the FQHE, particularly useful at even-denominator fillings such as $\nu = 1/2$, is provided by composite fermions [19]. Here each electron binds an even number $2p$ of magnetic flux quanta, forming an emergent fermionic quasiparticle. This flux attachment can be implemented formally by a singular gauge transformation, giving a mean-field description in which instead of electrons, we consider composite fermions moving in a reduced effective field

$$B_{\text{eff}} = B - 2p\phi_0 n_e, \tag{1.11}$$

where n_e is the electron density and $\phi_0 = h/e$ the flux quantum. At filling fractions

$$\nu = \frac{n}{2pn \pm 1}, \tag{1.12}$$

known as the Jain sequence, composite fermions fill n of their own effective Landau levels, so that the electronic FQHE is reinterpreted as an integer quantum Hall effect of composite fermions [19].

The Jain sequence converges, as $n \to \infty$, to $\nu = 1/(2p)$ — in particular $\nu = 1/2$ for $p = 1$ — where the attached flux exactly cancels the external field, $B_{\text{eff}} \to 0$. Rather than filling a well-defined integer number of effective Landau levels, the composite fermions here form a compressible, metallic state known as the composite Fermi liquid (CFL). This is visible in Fig. 1.4 as a clear *absence* of a Hall plateau at $\nu = 1/2$.

1.2.5 Higher Landau levels and non-abelian states

Non-abelian topological states are, in principle, possible already in the lowest Landau level, but have proven largely unattainable there for Coulomb interactions [48]. They do appear, however, in higher Landau levels, most notably as the prominent Hall plateau at $\nu = 5/2$ (Fig. 1.4) [52], for which the Moore–Read wavefunction was proposed as an explanation [32]. Like the Laughlin state, the Moore–Read state is the exact ground state of a model Hamiltonian — here, one containing a three-body interaction term — and has been shown numerically to describe the true Coulomb ground state at $\nu = 5/2$ well. It is however not particularly robust, lying close to competing phases, and fine-tuning of the interaction is needed to stabilize it [44]. Note that the filling $\nu = 5/2$ does not correspond to half filling of the $n = 2$ Landau level, but rather to the lowest Landau level being filled by two spin species while the $n = 1$ level is half filled by one of them.

Because the Moore–Read state can be understood as a p-wave paired state of composite fermions, its ground-state degeneracy depends characteristically on the parity of the particle number. For a genus-g manifold, this degeneracy is $3(M + 1)^g$ for an even number of fermions and $(M + 1)^g$ for an odd number, where M is defined through the filling fraction $\nu = 1/(M + 1)$ [3]. For the states studied later in this thesis, $g = 1$ (torus, periodic boundary conditions) and $M = 1$ (i.e., $\nu_{n=1} = 1/2$), giving expected degeneracies of 6 and 2 for even and odd particle number, respectively [36, 40].

1.2.6 Spectral flow

Additional magnetic flux can be threaded through either of the two independent non-contractible cycles of the torus on which a quantum Hall system is defined. Such flux insertion modifies the effective boundary conditions used to solve for the single-particle eigenstates, and hence the many-body spectrum. Inserting an integer number of flux quanta corresponds to a large gauge transformation, under which the Hamiltonian is invariant, so the spectrum must return to itself. This, however, does *not* imply that the eigenstates themselves are unchanged: adiabatic flux insertion acts on the degenerate ground-state manifold in the same way as a magnetic translation of all particles by one lattice vector, cyclically permuting the states rather than leaving them fixed — a phenomenon known as spectral flow [42].

Together with the ground-state degeneracy, spectral flow is used as a numerical diagnostic of topological order, provided the spectral gap remains open in the thermodynamic limit [42]. Both diagnostics are used extensively in Chapters 4 and 5 to identify candidate topological ground states in the finite-size numerics.

1.3 Topology of moiré materials

Despite its topological character, the quantum Hall effect, as described above, requires a strong external magnetic field, which is both impractical for applications and raises the conceptual question of whether Landau quantization is in fact essential to topological band structure. This motivated the search for topological states realizable in two-dimensional systems *without* a net magnetic field. The first lattice model to achieve this, the Haldane model [18], broke time–reversal symmetry explicitly through a complex next-nearest-neighbor hopping term acting as an analogue of the Aharonov–Bohm phase. Over a range of parameters, the resulting band structure acquires a nonzero Chern number. This model was later adapted to a more realistic setting in graphene [20], where the same complex hopping, arising directly from intrinsic spin-orbit coupling, plays the role of an effective, spin-dependent magnetic field built into the band structure itself.

A topologically nontrivial phase with nonzero Chern number that reproduces the integer or fractional quantum Hall effect without an external field is called a Chern insulator (CI) or

fractional Chern insulator (FCI), respectively [34, 42, 46]. Both explicitly break time-reversal symmetry. A time-reversal-invariant analogue can be built by combining two time-reversed copies of a Chern insulator with opposite Chern numbers: the total Chern number vanishes and time-reversal symmetry is restored, so the net charge Hall conductance is zero. If the spin projection S_z remains a good quantum number, the two spin sectors decouple into independent Chern insulators of opposite Chern number, and the system exhibits a spin quantum Hall effect, with opposite spin channels carrying counter-propagating, oppositely signed currents; this is already an example of a topological insulator (TI), classified by a $\mathbb{Z}_2$ invariant. Conservation of S_z is not, however, a requirement for this topological classification: even when spin-orbit coupling breaks S_z conservation and the spin quantum Hall effect no longer strictly applies, the $\mathbb{Z}_2$ invariant remains well defined, so the system can remain a TI. The interacting, fractionally filled analogue of the TI — built from two time-reversed copies of an FCI rather than a CI — is called a fractional topological insulator (FTI).

An analogous mechanism operates in TMDs. As discussed in §1.1.1, the low-energy physics of a group-VI TMD monolayer near K and $-K$ is described by a massive Dirac Hamiltonian,

$$\hat{H}(\mathbf{k}) = \mathbf{\Delta}(\mathbf{k}) \cdot \hat{\sigma}, \tag{1.13}$$

following Xiao et al. [54]. This is formally analogous to a two-level system in a momentum-dependent effective "magnetic field" $\mathbf{\Delta}(\mathbf{k})$, here generated by strong spin-orbit coupling. Because the Equation (1.13) is parametrized by momentum, the machinery of §1.2.2 applies: writing $\mathbf{R} = \mathbf{k} = [k_x, k_y]$, the Chern number of each eigenstate follows from Eq. (1.8). Depending on the sign of the effective mass term, the valence band acquires Chern number $C = \pm 1$ in the K and $-K$ valleys, respectively. In a full multi-band continuum model the global band topology depends strongly on the twist angle: for $MoTe_2$ near $\theta \approx 1.89°$, the two highest occupied bands together with the lowest conduction band all carry $C = 1$ and, combined with their flat dispersion, closely emulate the physics of the three lowest Landau levels [28, 55].

Since the approach adopted in this thesis models the moiré bands as Landau levels in effective magnetic fields, we use the terminology IQH/CI and FQH/FCI interchangeably when referring to the corresponding integer and fractional topological phases.

1.3.1 Valley polarization and inter-valley coherence

Since the two valleys carry a conserved $U(1) \times U(1)$ charge (§1.1.1), it is natural to treat the valley index τ as an internal degree of freedom, in analogy with real spin. In this analogy, it is easier to see that the system can order in two qualitatively different ways: by polarizing along the "Ising" axis, populating one valley preferentially over the other (valley polarization), or by ordering in the "XY" plane, developing a fixed phase relationship between the two valleys. The latter corresponds to a nonzero expectation value of the off-diagonal, inter-valley density operator, $\langle d^\dagger_{k,+} d_{k',-} \rangle \neq 0$, and defines an inter-valley-coherent (IVC) state. An IVC state breaks the emergent $U(1) \times U(1)$ valley symmetry down to the diagonal $U(1)$ of total charge. IVC order has been found in Hartree–Fock treatments of moiré model on which the thesis is based [23].

1.4 Goals of the thesis

Although moiré materials and Landau level models have been predicted and experimentally shown to host a variety of topological phases, understanding their behavior under realistic, tunable experimental conditions remains challenging. This is particularly true for the time-reversal-invariant topological phases, fractional topological insulators (FTIs): they are the subject of ongoing theoretical research [23], with first candidate experimental signatures reported only very recently [51], while the non-abelian FTIs — particularly relevant for applications — have so far been studied only theoretically [1]. The goal of this thesis is to shed light on how the

moiré potential and dielectric screening influence the stability of topological phases, focusing on abelian and non-abelian FTIs, using a model that is simple enough for systematic numerical study while remaining relevant to the experimentally dominant moiré TMD systems.

The remainder of this thesis investigates moiré materials numerically via the valley-as-Landau-level correspondence, using the exact diagonalization framework of Chapter 3. Concretely, we study the two-valley Landau-level model of Chapter 2, including tunable bandwidth and inter–valley coupling. We extend the model to include inverted single-particle dispersion between the two valleys, and projection onto higher Landau levels.

In Chapter 4, three fillings expected to host distinct phases are examined: $\nu = 2/3$, where a valley-polarized fractional quantum Hall state and a valley-unpolarized fractional topological insulator are both expected to compete with gapless states; $\nu = 1$, where the analogous integer quantum Hall and composite-Fermi-liquid phases are expected; and $\nu = 1/2$, included as a reference filling with no expected gapped topological order. At each filling, we map out the phase diagram as a function of the inter-valley coupling α and the single-particle bandwidth W, additionally examining the screening parameters d_{SC} and l_{RK}, using ground-state degeneracy, spectral flow under flux insertion, and momentum-space occupation as diagnostics. Some of the resulting numerical phase boundaries are further qualitatively compared against analytic Hartree–Fock estimates of the valley-depolarization transition, providing an independent check on the numerics and on the transition's dependence on the model's screening parameters.

Chapter 5 extends the calculation to the first excited Landau level, focusing on the filling $\nu = 3$ ($\nu_{n=1} = 1$ or $\nu_{n=1} = 1/2$ per valley), where the valley-unpolarized phase is expected to host a non-abelian fractional topological insulator. The corresponding phase diagram is mapped out, the stability of the FTI is discussed, and dependence of its extent on screening parameters is studied.

Chapter 2

Model

The model used here, based on the work of Kwan et al. [23], is designed to capture the essential physics of moiré materials. As motivated in §1.3, the two valleys of a moiré TMD carry opposite Chern number and, near the relevant magic angle, closely emulate a pair of oppositely oriented Landau levels; we take this correspondence as the starting point of the model. For simplicity of implementation and interpretation, we work on a square lattice with a lattice constant $a = \sqrt{2\pi}$, with the magnetic length $l_B = 1$ corresponding to one flux quantum per unit cell. The magnitude of the primitive reciprocal lattice vector is then $Q = 2\pi/a = a = \sqrt{2\pi}$. The single-particle Hilbert space consists of two Landau levels carrying a valley index $\tau = \pm 1$, which experience opposite effective magnetic fields $\mathbf{B}_\tau = -\tau B\hat{\mathbf{z}}$, corresponding to opposite Chern numbers $C = \tau$. Single-particle states within each Landau level are given by magnetic Bloch functions, eigenfunctions of the magnetic translation operators labeled by momentum $\mathbf{k}$ in the magnetic Brillouin zone. We denote by $\hat{d}^\dagger_{\mathbf{k},\tau}$ the operator creating a particle in the corresponding state. We work on a torus (periodic boundary conditions) and treat the magnetic vector potential in the periodic gauge. To represent a generic moiré material, the model combines a screened Coulomb interaction projected onto a single Landau level (i.e., neglecting Landau-level mixing) with a single-particle potential representing that of a real material. Relative to the original formulation of Ref. [23], the model is modified to allow projection onto higher Landau levels, and to allow for an inverted single-particle dispersion in one of the two valleys.

2.1 Interactions

The interaction is modeled as a screened Coulomb density-density interaction projected onto the Landau level, with a tunable inter-valley coupling strength:

$$H_{\text{int}} = \frac{1}{2A} \sum_{\substack{\mathbf{k}\in\text{BZ} \\ \mathbf{k}'\in\text{BZ} \\ \mathbf{q}}} \tilde{U}_{\tau\tau'}(\mathbf{q})\, \Lambda_\tau(\mathbf{k},\mathbf{q})\, \Lambda_{\tau'}(\mathbf{k}',-\mathbf{q})\, d^\dagger_{\mathbf{k},\tau}\, d^\dagger_{\mathbf{k}',\tau'}\, \hat{d}_{\lfloor \mathbf{k}'-\mathbf{q}\rfloor,\tau'}\, \hat{d}_{\lfloor \mathbf{k}+\mathbf{q}\rfloor,\tau}, \tag{2.1}$$

where A is the area of the system, τ, τ' is the valley index and

$$\tilde{U}_{\tau\tau'}(\mathbf{q}) = \frac{2\pi}{q(1 + l_{\text{RK}} q)} \tanh(q\, d_{\text{SC}})\, (1 + \alpha\, \tau\tau') \tag{2.2}$$

is the interaction potential. The $l_{\text{RK}} = 0$ limit of Eq. (2.2) is the screened Coulomb interaction of a charge held midway between two perfectly conducting plates separated by a distance $2d_{\text{SC}}$. The Rytova–Keldysh correction $l_{\text{RK}} \neq 0$ additionally accounts for the dielectric environment and the polarizability of the sample itself [11, 21, 45]. The parameter $\alpha \in [-1, 1]$ controls the relative strength of inter- and intra-valley interactions and is central to the calculations that follow: at $\alpha = 0$ the interaction is valley-isotropic with $\tilde{U}_{+,-} = \tilde{U}_{-,+} = \tilde{U}_{+,+} = \tilde{U}_{-,-}$; at

$\alpha = 1$ only the intra-valley interaction survives, $\tilde{U}_{+,-} = \tilde{U}_{-,+} = 0$; and at $\alpha = -1$ only the inter-valley interaction survives. While it is postulated that α can vary in real materials due to band mixing [1], the $\alpha \approx 0$ limit is considered most experimentally relevant.

The interaction is projected onto a Landau level through the form factor Λ, which, in the periodic gauge with $\mathbf{k} \in [0, Q) \times [0, Q)$ (the Brillouin zone), takes the following form for the lowest Landau level:

$$\Lambda_\tau(\mathbf{k}, \mathbf{q}) = e^{-|\mathbf{q}|^2/4}\, e^{-i\tau(-q_x(k_y+q_y/2) - k_x k_y + \lfloor k_x+q_x \rfloor (k_y+q_y))}, \tag{2.3}$$

where the operator $\lfloor \cdot \rfloor$ denotes reduction modulo Q, mapping momenta back into the first Brillouin zone.

Alternatively, the interaction can be expressed in a smooth gauge, with $\hat{d}_{\lfloor k \rfloor} \to \hat{d}_k$ and the form factor:

$$\Lambda_\tau(\mathbf{k}, \mathbf{q}) = e^{-|\mathbf{q}|^2/4}\, e^{i\tau(k_x+q_x/2)q_y}. \tag{2.4}$$

Whereas the periodic gauge (2.3) is convenient for numerical calculations as it enforces periodic boundary conditions, the smooth gauge (2.4) is more suitable for analytical treatments, as below in §2.3.

Abouelkomsan and Fu [1] further showed that projecting the interaction onto higher Landau levels amounts, in this formalism, to multiplying the lowest-Landau-level form factor by the Laguerre polynomial L_n of the corresponding Landau-level index n:

$$\Lambda_{n,\tau}(\mathbf{k}, \mathbf{q}) = \Lambda_\tau(\mathbf{k}, \mathbf{q})\, L_n\left(\frac{|\mathbf{q}|^2}{2}\right). \tag{2.5}$$

The two Laguerre polynomials relevant to this work are $L_0(x) = 1$ and $L_1(x) = 1 - x$, corresponding respectively to the lowest Landau level (LLL) and the first excited Landau level ($n = 1$ LL).

Throughout this work we consider two variants of the interaction potential $\tilde{U}_{\tau\tau'}(\mathbf{q})$ defined in Eq. (2.2). For most of the thesis, the full interaction is treated, as given in the equation above, matching Kwan et al. [23]. This is equivalent to considering the system without a uniform neutralizing background, or with one that couples to the total electron density (all energy levels shifted by a constant). It is denoted as $U(\mathbf{q} = 0) \neq 0$ in the following chapters. Zou, Li and MacDonald [59], however, argue that the physically relevant neutralizing background would be one where coupling to electron density scales with inter- and intra-valley interaction strength, giving a complete cancellation of the $\mathbf{q} = 0$ term of interaction, $\tilde{U}_{\tau\tau'}(\mathbf{q} = 0) = 0$ for all τ, τ'. The phase diagrams of this variant are also sketched for the LLL, denoted for simplicity as $U(\mathbf{q} = 0) = 0$. It is, however, worth noting that in the most physically relevant limit of $\alpha = 0$ both versions of the interaction are equivalent.

2.2 Single-particle dispersion

The single-particle dispersion is modeled as a lowest-harmonic cosine potential, compatible with the choice of the magnetic unit cell, following Kwan et al. [23] and Bultinck et al. [4]:

$$\epsilon_\tau(\mathbf{k}) = -\frac{W}{4}\left(\cos(k_x a) + \cos(k_y a)\right) \cdot \tau^{(1-\mathrm{sgn}(W))/2}. \tag{2.6}$$

Relative to these references, we introduce the valley-dependent prefactor $\tau^{(1-\mathrm{sgn}(W))/2}$, which allows for an inverted dispersion in one of the two valleys. For $W > 0$ the prefactor is $\tau^0 = 1$ and Eq. (2.6) reduces to the same dispersion in both valleys, as in the original model; for $W < 0$ the prefactor becomes $\tau^1 = \tau$, so that the dispersion is inverted in one of the valleys relative to the other.

2.3 Extension of the model to $W < 0$

Instead of considering the extension of the model as an algebraic modification of the dispersion relation, it can also be done through a unitary transformation of the many-body Hamiltonian. A consequence of this is that the phase diagram must be symmetric under $W \to -W$ in the thermodynamic limit. Let us dissect the Hamiltonian of Ref. [23] into parts,

$$\hat{H}(W) = \hat{H}^{+}_{\text{kin}}(W) + \hat{H}^{-}_{\text{kin}}(W) + \hat{H}_{\text{int}}, \tag{2.7}$$

where

$$\hat{H}^{\pm}_{\text{kin}}(W) = \sum_{\mathbf{k}} \epsilon_{\pm}(\mathbf{k}) \hat{d}^{\dagger}_{\mathbf{k},\pm} \hat{d}_{\mathbf{k},\pm} \tag{2.8}$$

is the kinetic energy in valley $\pm$ and $\hat{H}_{\text{int}}$ is the interaction term.

One can consider a unitary operator $\hat{C}$ such that

$$\hat{C}\hat{H}(W)\hat{C}^{-1} = \hat{H}(-W), \tag{2.9}$$

which is equivalent to

$$\hat{C}\hat{H}^{-}_{\text{kin}}(W)\hat{C}^{-1} = \hat{H}^{-}_{\text{kin}}(-W), \tag{2.10}$$

and identity operator on all other parts of the Hamiltonian. Effectively, this amounts to shifting (boosting) a chosen valley's $(-)$ dispersion by half the Brillouin zone, $\mathbf{Q}_0 = (Q/2, Q/2)$, since $\cos(k_i + Q/2) = -\cos(k_i)$, $i = x, y$.

Thus, the form of the operation $\hat{C}$ is deduced to be

$$\hat{C}\, \hat{d}^{\dagger}_{\mathbf{k},-}\, \hat{C}^{-1} = e^{i\Theta(\mathbf{k})}\, \hat{d}^{\dagger}_{\mathbf{k}+\mathbf{Q}_0,-}, \tag{2.11}$$

where $\Theta(\mathbf{k})$ is a gauge dependent phase factor. In the smooth gauge, its (non-unique) form follows from the basis operators satisfying the magnetic periodic boundary conditions:

$$\hat{d}^{\dagger}_{\mathbf{k}+Q\hat{x},\tau} = e^{i\tau k_y Q}\, \hat{d}^{\dagger}_{\mathbf{k},\tau}, \qquad \hat{d}^{\dagger}_{\mathbf{k}+Q\hat{y},\tau} = \hat{d}^{\dagger}_{\mathbf{k},\tau}. \tag{2.12}$$

The phase $\Theta(\mathbf{k})$ can be fixed by additionally requiring that two applications of $\hat{C}$ return the system to its original state,

$$\begin{aligned} \hat{C}^2\, \hat{d}^{\dagger}_{\mathbf{k},-}\, \hat{C}^{-2} &= e^{i\Theta(\mathbf{k})} e^{i\Theta(\mathbf{k}+\mathbf{Q}_0)}\, \hat{d}^{\dagger}_{\mathbf{k}+2\mathbf{Q}_0,-} \\ &= e^{i\Theta(\mathbf{k})} e^{i\Theta(\mathbf{k}+\mathbf{Q}_0)}\, e^{-ik_y Q}\, \hat{d}^{\dagger}_{\mathbf{k},-} \overset{!}{=} \hat{d}^{\dagger}_{\mathbf{k},-}, \end{aligned} \tag{2.13}$$

which is solved by the linear ansatz

$$\Theta(\mathbf{k}) = \frac{k_y Q}{2} + \Theta_0. \tag{2.14}$$

This informed guess for the operator $\hat{C}$ correctly transforms the many-body Hamiltonian, as we verify below. First, note that

$$\hat{C}\, \hat{d}^{\dagger}_{\mathbf{k},-} \hat{d}_{\mathbf{k}+\mathbf{q},-}\, \hat{C}^{-1} = \hat{C}\, \hat{d}^{\dagger}_{\mathbf{k},-}\, \hat{C}^{-1}\, \hat{C}\, \hat{d}_{\mathbf{k}+\mathbf{q},-}\, \hat{C}^{-1} = e^{i\frac{k_y Q}{2}}\, \hat{d}^{\dagger}_{\mathbf{k}+\mathbf{Q}_0,-} \hat{d}_{\mathbf{k}+\mathbf{q}+\mathbf{Q}_0,-}. \tag{2.15}$$

Action on the kinetic energy. By definition, $\hat{C}$ acts only on the $-$ valley. Using Eq. (2.15) with $\mathbf{q} = 0$,

$$\begin{aligned}
\hat{C}\,\hat{H}^{-}_{\mathrm{kin}}(W)\,\hat{C}^{-1} &= -\frac{W}{4}\sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon(\mathbf{k})\,\hat{C}\,\hat{d}^{\dagger}_{\mathbf{k},-}\hat{d}_{\mathbf{k},-}\,\hat{C}^{-1} \\
&= -\frac{W}{4}\sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon(\mathbf{k})\,\hat{d}^{\dagger}_{\mathbf{k}+\mathbf{Q}_0,-}\hat{d}_{\mathbf{k}+\mathbf{Q}_0,-} \\
&= -\frac{W}{4}\sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon(\mathbf{k}-\mathbf{Q}_0)\,\hat{d}^{\dagger}_{\mathbf{k},-}\hat{d}_{\mathbf{k},-} \\
&= -\frac{W}{4}\sum_{\mathbf{k}'\in\mathrm{BZ}} \big[-\epsilon(\mathbf{k}')\big]\,\hat{d}^{\dagger}_{\mathbf{k}',-}\hat{d}_{\mathbf{k}',-} \\
&= \hat{H}^{-}_{\mathrm{kin}}(-W),
\end{aligned} \tag{2.16}$$

where the limits of summation are left unchanged owing to the periodicity of the summands modulo the Brillouin zone. The operation $\hat{C}$ therefore correctly transforms the kinetic energy of the $-$ valley.

Action on the LL density operators. Using Eq. (2.15) (generalized to valley τ),

$$\begin{aligned}
\hat{C}\,\hat{\rho}_{\tau}(\mathbf{q})\,\hat{C}^{-1} &= \sum_{\mathbf{k}\in\mathrm{BZ}} \Lambda_{\tau}(\mathbf{k},\mathbf{q})\,\hat{C}\,\hat{d}^{\dagger}_{\mathbf{k},\tau}\,\hat{C}^{-1}\,\hat{C}\,\hat{d}_{\mathbf{k}+\mathbf{q},\tau}\,\hat{C}^{-1} \\
&= e^{i\frac{q_y Q}{2}} \sum_{\mathbf{k}\in\mathrm{BZ}} \Lambda_{\tau}(\mathbf{k},\mathbf{q})\,\hat{d}^{\dagger}_{\mathbf{k}+\mathbf{Q}_0,\tau}\hat{d}_{\mathbf{k}+\mathbf{q}+\mathbf{Q}_0,\tau} \\
&= e^{i\frac{q_y Q}{2}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \Lambda_{\tau}(\mathbf{k}'-\mathbf{Q}_0,\mathbf{q})\,\hat{d}^{\dagger}_{\mathbf{k}',\tau}\hat{d}_{\mathbf{k}'+\mathbf{q},\tau} \\
&= e^{i\frac{q_y Q}{2}}\,e^{-i\frac{q_y Q}{2}} \sum_{\mathbf{k}'\in\mathrm{BZ}} \Lambda_{\tau}(\mathbf{k}',\mathbf{q})\,\hat{d}^{\dagger}_{\mathbf{k}',\tau}\hat{d}_{\mathbf{k}'+\mathbf{q},\tau} \\
&= \hat{\rho}_{\tau}(\mathbf{q}).
\end{aligned} \tag{2.17}$$

Since the LL density operator is invariant under $\hat{C}$, the interaction term is invariant as well.

Thus we have seen that the operation $\hat{C}$ correctly transforms the many-body Hamiltonian of Ref. [23] with $W > 0$ into the one with $W < 0$. Consequently, there is a bijective mapping between the many-body states of the two Hamiltonians, and the phase diagrams should be symmetric with respect to the sign of W — up to the difference in sampling of Brillouin zone points under the momentum shift by $\mathbf{Q}_0$, which is irrelevant in the thermodynamic limit.

2.3.1 Symmetry of the $W < 0$ Hamiltonian

It is worth noting that while the $W > 0$ Hamiltonian is time-reversal symmetric, the $W < 0$ Hamiltonian is not. However, a new symmetry can be constructed from $\hat{C}$ and the time-reversal operator $\hat{T}$. Starting from the time-reversal symmetry of $\hat{H}(W)$,

$$\hat{T}\hat{H}(W)\hat{T}^{-1} = \hat{H}(W), \tag{2.18}$$

we conjugate both sides by $\hat{C}$ and insert $\hat{C}^{-1}\hat{C} = \mathbb{1}$:

$$\begin{aligned}
\hat{C}\,\hat{T}\hat{H}(W)\hat{T}^{-1}\,\hat{C}^{-1} &= \hat{C}\,\hat{H}(W)\,\hat{C}^{-1}, \\
\hat{C}\hat{T}\,(\hat{C}^{-1}\hat{C})\,\hat{H}(W)\,(\hat{C}^{-1}\hat{C})\,\hat{T}^{-1}\hat{C}^{-1} &= \hat{C}\,\hat{H}(W)\,\hat{C}^{-1}, \\
(\hat{C}\hat{T}\hat{C}^{-1})\,\hat{H}(-W)\,(\hat{C}\hat{T}^{-1}\hat{C}^{-1}) &= \hat{H}(-W).
\end{aligned} \tag{2.19}$$

Thus, the $W < 0$ Hamiltonian is symmetric under the operation $\hat{C}\hat{T}\hat{C}^{-1}$.

Chapter 3

Methods

Having introduced the model Hamiltonian in Chapter 2, we now describe how it is solved numerically. All results in Chapter 4 and 5 are obtained by exact diagonalization (ED) on finite tori, using the criteria for identifying topological order laid out at the end of this chapter.

3.1 Exact diagonalization

The model Hamiltonian is solved using the DiagHam package for exact diagonalization [41]. On a torus geometry (periodic boundary conditions), the center-of-mass (COM) momentum is a good quantum number [3], which allows an initial block-diagonalization of the Hamiltonian into sectors labeled by the COM momentum $\mathbf{K} = (K_x, K_y)$, and the valley polarization $P_z = N_+ - N_-$ (N_τ being the occupation of the τ valley). Within each block, the Hamiltonian matrix is constructed explicitly, and its lowest few eigenvalues are found iteratively using the Lanczos algorithm [25]. The number of eigenvalues requested in each sector is chosen to exceed the maximum ground-state degeneracy expected for the filling under consideration.

3.2 Momentum grid discretization and tilted boundary conditions

To construct the finite Hamiltonian matrix, the Brillouin zone $[0, Q) \times [0, Q)$ must be discretized into a finite set of momentum points. This grid is not unique: as shown by Läuchli et al. [26] and Repellin et al. [43] in the context of fractional Chern insulators, different admissible choices of a real-space cluster correspond to very different sampling of the same Brillouin zone, and this choice can noticeably affect the stability of the resulting topological phases.

The grid is constructed by first choosing two real-space spanning vectors,

$$\mathbf{T}_1 = n_{x1}\mathbf{a}_1 + n_{y1}\mathbf{a}_2, \qquad \mathbf{T}_2 = n_{x2}\mathbf{a}_1 + n_{y2}\mathbf{a}_2, \tag{3.1}$$

with integer coefficients $n_{x1}, n_{y1}, n_{x2}, n_{y2}$, where $\mathbf{a}_1, \mathbf{a}_2$ are the primitive lattice vectors of the underlying square lattice. These two vectors define the finite cluster on which the many-body problem is solved, enclosing $N_\phi = |n_{x1}n_{y2} - n_{x2}n_{y1}|$ unit cells (flux quanta).

Invariance of the many-body wavefunction under a translation by $\mathbf{T}_1$ or $\mathbf{T}_2$ forces the single-particle momentum $\mathbf{k}$ to satisfy $\mathbf{k} \cdot \mathbf{T}_1, \mathbf{k} \cdot \mathbf{T}_2 \in 2\pi\mathbb{Z}$. Equivalently, $\mathbf{k}$ must lie on the lattice spanned by the reciprocal vectors $\tilde{\mathbf{T}}_1, \tilde{\mathbf{T}}_2$ dual to $\mathbf{T}_1, \mathbf{T}_2$,

$$\tilde{\mathbf{T}}_i \cdot \mathbf{T}_j = 2\pi\delta_{ij}, \qquad i, j = 1, 2. \tag{3.2}$$

The admissible momenta are the N_ϕ points of this reciprocal lattice that fall within one Brillouin zone, i.e. they are defined modulo $\tilde{\mathbf{a}}_1$ and $\tilde{\mathbf{a}}_2$ (reciprocal vectors dual to $\mathbf{a}_1, \mathbf{a}_2$: $\tilde{\mathbf{a}}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$).

Concretely, we label them by two integers (p, q),

$$\mathbf{k}_{p,q} = p\,\tilde{\mathbf{T}}_1 + q\,\tilde{\mathbf{T}}_2 \pmod{\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2}, \tag{3.3}$$

choosing $p \in [0, N_1)$ and $q \in [0, N_2)$ with $N_1 N_2 = N_\phi$, such that every point is counted exactly once and no two choices of (p, q) give equivalent momenta. In practice this means imposing $N_i \tilde{\mathbf{T}}_i \equiv 0 \pmod{\tilde{\mathbf{a}}_1, \tilde{\mathbf{a}}_2}$, $i \in \{1, 2\}$, and wrapping $\mathbf{k}_{p,q}$ back into the first BZ whenever it falls outside of it.

For a fixed N_ϕ, this construction allows much more freedom in choosing the cluster shape than an untilted grid. Läuchli et al. [26] mention that clusters with $|\mathbf{T}_1|/|\mathbf{T}_2| \gg 1$ tend to favor charge-density-wave states over the target topological phase; this was the main criterion used here for selecting the tilting, i.e. $\mathbf{T}_1, \mathbf{T}_2$ were chosen so that $|\mathbf{T}_1|/|\mathbf{T}_2|$ is as close to 1 as possible. Among the remaining choices with comparable aspect ratio, the angle between $\mathbf{T}_1$ and $\mathbf{T}_2$ was further chosen to be as close as possible to 60° (equivalently 120°), so that the resulting momentum-space sampling is most isotropic, i.e. closest to a uniform hexagonal grid.

For simplicity, however, unless stated otherwise, the calculations presented in this thesis use the conventional (untilted) boundary conditions $\mathbf{T}_i = N_i \mathbf{a}_i$, corresponding to a rectangular grid of momenta in the Brillouin zone. Tilted boundary conditions are used only where explicitly noted, primarily for the momentum-occupation calculations of Chapter 4, and in Chapter 5, where less stable non-abelian phases are expected.

3.3 Twisted boundary conditions and flux insertion

To explicitly check for spectral flow, twisted boundary conditions are used, whereby an additional flux $\mathbf{\Phi} = (\Phi_1, \Phi_2)$ is inserted into the system through the two cycles of the torus. Because the two valleys experience opposite effective magnetic field (and carry opposite Chern number $C = \tau$), the flux is inserted with opposite sign in the two valleys,

$$\mathbf{\Phi}_\tau = \tau\,\mathbf{\Phi}. \tag{3.4}$$

This is equivalent to an effective uniform, valley-dependent shift of the momentum grid sampled in the Hamiltonian,

$$\mathbf{k}_\tau = \mathbf{k}^0 + \tau \frac{\Phi_1}{2\pi} \tilde{\mathbf{T}}_1 + \tau \frac{\Phi_2}{2\pi} \tilde{\mathbf{T}}_2 \tag{3.5}$$

where $\mathbf{k}^0$ is the original (untwisted) momentum. Here, $\Phi_i = 2\pi$ corresponds to one flux quantum — it shifts the momentum grid by exactly one grid step in the corresponding direction, which further shows that the spectrum must be identical at $\Phi_i = 0$ and $\Phi_i = 2\pi$: the twisted grid coincides with the original one, merely relabeled.

This valley-antisymmetric choice is what makes flux insertion compatible with the symmetries of the Hamiltonian: time-reversal symmetry $\hat{T}$ at $W > 0$ and the combined symmetry $\hat{C}\hat{T}\hat{C}^{-1}$ of §2.3 at $W < 0$.

Throughout the thesis the flux is inserted along $\tilde{\mathbf{T}}_1$ ($\Phi_2 = 0$), which corresponds to the x axis for the untilted boundary conditions.

3.4 Numerical identification of topological phases

The nature of each ground-state phase is determined from the following characteristics.

First, the ground-state valley polarization is identified as the P_z sector containing the lowest-energy eigenstate.

Second, the gaps between consecutive energy levels are computed. Exact diagonalization is performed for 10 equally spaced flux values $\phi \in [0, 2\pi]$. For a candidate N-fold degenerate

ground-state manifold, we define the spectral gap as

$$\Delta_N = \begin{cases} \min_\phi E_{N+1}(\phi) - \max_\phi E_N(\phi), & \text{if } \{\mathbf{k}_n(\phi_1)\}_{n=1}^N = \{\mathbf{k}_n(\phi_2)\}_{n=1}^N \ \ \forall\, \phi_1, \phi_2 \in [0, 2\pi], \\ 0 & \text{otherwise.} \end{cases} \tag{3.6}$$

The gap is evaluated for every N relevant to the filling under consideration. Here, $E_N(\phi)$ denotes the energy of the N-th state at flux ϕ, with the spectrum ranked across all center-of-mass momentum and valley-polarization sectors, and $\mathbf{k}_n(\phi)$ denotes the center-of-mass momentum of that state. The spectral gap is therefore determined by the separation between the highest energy attained by the candidate ground-state manifold and the lowest energy attained by the first excited state over the full flux interval. This ensures that the entire N-fold manifold remains separated from higher-energy states throughout flux insertion. The second condition sets the gap to zero if the set of COM momenta spanning the candidate ground-state manifold changes with flux, which signals a level crossing between the candidate manifold and a state outside it and hence a closing of the gap.

Third, the variation of spectra under flux insertion is inspected visually for spectral flow between the states within the ground-state manifold.

As discussed in Chapter 1, spectral flow, ground-state degeneracy, and the absence of gap closing in the thermodynamic limit together constitute strong evidence for a gapped topological phase. Thus, additionally to probes above, a fully rigorous identification would require finite-size scaling. Due to the limited computational resources, such a systematic scaling analysis was not performed; the results presented here should therefore be understood as consistent with, rather than a definitive proof of, the identified phases.

Finally, for some systems the momentum-space occupation $n(\mathbf{k}) = |\langle \mathbf{k}|\psi_0\rangle|^2$ of the ground state at zero flux is additionally computed, in order to distinguish the flat occupation characteristic of topological states (e.g. FCIs) or of gapless composite-Fermi-liquid-like states from the peaked occupation expected of a Fermi liquid.

The description of numerical criteria used for each phase is deferred until after the discussion of suspected phases at different fillings, and is done in Table 4.1 for the LLL and Table 5.1 for the $n = 1$ LL.

It is also worth noting that due to the lack of inter-valley tunneling in the model, the inter-valley coherence order parameter $\langle d^\dagger_{k,+} d_{k',-}\rangle$ is identically zero in all calculations, and hence cannot be directly used.

Chapter 4

Results — lowest Landau level

We now turn to results for the lowest Landau level. Phase-diagram calculations (ground-state valley polarization, spectral gaps, and flux insertion) are performed on a 3×4 system with the default (untilted) Brillouin-zone sampling, scanning W and α. The momentum-occupation calculations are instead performed on a tilted lattice, with the Brillouin-zone sampling shown in Fig. 4.13c. Unless stated otherwise, $d_{\mathrm{SC}} = 6a$ and $l_{\mathrm{RK}} = 0$ are used as default parameters, following Kwan et al. [24].

To connect with previous theoretical and experimental work, and because they are the fillings at which topological phases are expected, we consider $\nu = \frac{1}{2}$, $\nu = \frac{2}{3}$, and $\nu = 1$.

4.1 Anticipated phases

4.1.1 $\nu = \frac{2}{3}$ ($\nu = \frac{1}{3}$ per valley)

At the decoupled limit $\alpha = 1$, $W = 0$, the intra-valley repulsion favors a valley-unpolarized $\frac{1}{3}+\frac{1}{3}$ state, and the ground state is expected to be a fractional topological insulator (FTI) consisting of two time-reversal-conjugate copies of the Laughlin $\frac{1}{3}$ state, $|\mathrm{FTI}\rangle = |\mathrm{FQH}\rangle \otimes \overline{|\mathrm{FQH}\rangle}$ (with overline denoting time-reversal conjugate state), with a characteristic $3 \times 3 = 9$-fold ground-state degeneracy on the torus [8]. At low α, the large inter-valley interaction should instead favor a valley-polarized phase, with a $\nu = \frac{2}{3}$ Laughlin-type fractional quantum Hall (FQH) state expected in a single valley, consistent with prior theoretical work at the same filling [10]. Experimentally, fractional Chern insulator states have been reported in twisted $MoTe_2$ at closely related fillings [5, 38, 56, 57], supporting this general picture.

The nature of the states and transitions connecting the FQH and FTI phases as α is tuned is not well understood *a priori*. Near the decoupled limit $\alpha \to 1$, McDonald and Haldane [31] studied a closely related problem of two-component quantum Hall states at $\nu_T = 2/3$ using a K-matrix Chern–Simons description, and found that increasing the inter-component coupling drives a transition from the decoupled $(3, 3, 0)$-type state (equivalent to our FTI) to a competing $(1, 1, 2)$ Halperin state. However, contrary to the literature, our two valleys carry opposite Chern number — the standard holomorphic Halperin (m, n, l) wavefunction construction does not directly apply.

Independently, the bandwidth W is expected to compete against these interaction-favored ordered states: in the limit $W \to \infty$, kinetic energy favors a gapless, valley-unpolarized Fermi liquid ($\mathrm{FL}_{\mathrm{unpol.}}$) state, so transitions that destroy both the valley polarization and the topological order of the low–α phases (FTI, FQH) are expected. While the nature of transitions out of gapped topological states has not been addressed in the literature, the valley-depolarization transition is expected to be Stoner-like and was treated with a mean-field approach in Ref. [39]. That reference also notes that, depending on the inter-valley coupling, a similar Stoner mecha-

nism can instead drive an FL→inter-valley-coherent (IVC) transition rather than valley polarization; the IVC state is therefore also a candidate phase at intermediate α.

4.1.2 $\nu = 1$ ($\nu = \frac{1}{2}$ per valley)

At $\alpha = 1$, $W = 0$, the ground state is expected to be a pair of decoupled composite Fermi liquids, $|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$, consistent with exact-diagonalization studies of the continuum $MoTe_2$ model at the same filling [15].

At small α, valley polarization is again expected, giving an integer quantum Hall (IQH) state. Since it corresponds to integer filling of the LLL, the Hilbert space at full valley-polarization is one-dimensional, and no other state is possible in this region. As before, increasing $|W|$ is expected to drive a Stoner-like depolarization transition to an FL_u state. Consistently with the Stoner treatment of Ref. [39], an IVC state is additionally expected to appear at intermediate α, again transitioning to an $\mathrm{FL}_{\mathrm{unpol.}}$ state at large $|W|$. This IVC state was found directly in a Hartree–Fock treatment of the same model, in Ref. [23]. The transition between the IVC state and the decoupled CFL pair as α increases, however, has not been directly addressed in the literature.

The two-valley LLL and 1LL model was also studied by Das, Wagner, and Neupert [12]. By tuning the Haldane pseudopotentials, they found phases similar to the ones described above, in addition to partially particle–hole-transformed Halperin states.

4.1.3 $\nu = \frac{1}{2}$ ($\nu = \frac{1}{4}$ per valley)

We do not expect a gapped topological phase at this filling. By analogy with $\nu = 1$, a composite-Fermi-liquid state $|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$ is expected at $\alpha = 1$, and a single valley-polarized CFL (denoted by $|\mathrm{CFL}\rangle_{\mathrm{pol.}}$) is expected at low α in the valley-polarized region, with both giving way to a Fermi liquid at large $|W|$. We are not aware of prior literature directly addressing this specific filling.

4.1.4 Numerical criteria for defining phases

As the thesis focuses on gapped topological FQH and FTI phases, the main phase diagrams shown in this chapter use only a general distinction of phases based on valley polarization, spectral gap, and presence of spectral flow, for both $U(\mathbf{q} = 0) = 0$ and $U(\mathbf{q} = 0) \neq 0$ interactions. In this way, however, separate non-topological phases are not resolved and are covered under the umbrella labels "UP" (unpolarized) and "VP" (polarized), as shown in Table 4.1a. For $U(\mathbf{q} = 0) \neq 0$, these regions are described in more detail in the text and in the summary phase diagrams of Fig. 6.1, using indirect signatures ($n(\mathbf{k})$), known analytic limits, and past literature, rather than the gap/spectral-flow criteria of Table 4.1a. These candidate phases, listed in Table 4.1b, are consistent with the numerics but not directly confirmed by them.

4.2 Mean-field treatment of valley-depolarizing phase transitions

To understand the phase transitions out of the valley-polarized phase, we treat the interaction Hamiltonian in a Hartree–Fock mean-field approximation. Starting from

$$\hat{H}_{\mathrm{int}} = \frac{1}{2A} \sum_{\mathbf{k},\mathbf{k}',\mathbf{q}} \sum_{\tau,\tau'} \tilde{U}_{\tau\tau'}(\mathbf{q}) \Lambda_{\tau}(\mathbf{k},\mathbf{q}) \Lambda_{\tau'}(\mathbf{k}',-\mathbf{q})\, \hat{d}^{\dagger}_{\mathbf{k},\tau} \hat{d}^{\dagger}_{\mathbf{k}',\tau'} \hat{d}_{\mathbf{k}'-\mathbf{q},\tau'} \hat{d}_{\mathbf{k}+\mathbf{q},\tau}, \tag{4.1}$$

Wick's theorem gives two contraction channels for the quartic operator: a *direct* (Hartree) channel, pairing $\hat{d}^{\dagger}_{\mathbf{k},\tau}$ with $\hat{d}_{\mathbf{k}+\mathbf{q},\tau}$, and an *exchange* (Fock) channel, pairing $\hat{d}^{\dagger}_{\mathbf{k},\tau}$ with $\hat{d}_{\mathbf{k}'-\mathbf{q},\tau'}$.

(a) Primary phases

Filling	Label	P_z	Criterion
$\nu = \frac{2}{3}$	VP	full	gapless
	FQH	full	$\Delta_3 > 10^{-3}$, spectral flow
	UP	0	no spectral flow (unresolved)
	FTI	0	$\Delta_9 > 10^{-3}$, spectral flow
$\nu = 1$	IQH	full	$\Delta_1 > 0$
	UP	0	no spectral flow (unresolved)
$\nu = \frac{1}{2}$	VP	full	gapless (unresolved)
	UP	0	gapless (unresolved)

(b) Candidate phases in unresolved regions

Filling	Region	Regime	Candidate	Signature
$\nu = \frac{2}{3}$	VP	intermediate $\lvert W\rvert$	Fermi liquid $\lvert\mathrm{FL}\rangle_{\mathrm{pol.}}$	gapless, peaked $n(\mathbf{k})$
	UP	large $\lvert W\rvert$	Fermi liquid $\lvert\mathrm{FL}\rangle_{\mathrm{unpol.}}$	gapless, peaked $n(\mathbf{k})$
	UP	small $\lvert W\rvert$	CDW-like	$\Delta_{3/6} > 10^{-3}$, no spectral flow
	UP	small $\lvert W\rvert$, neighbors FCI	IVC	gapless, flat $n(\mathbf{k})$
$\nu = 1$	UP	untilted cluster	phase-separation	$\Delta_{12} > 10^{-3}$, no spectral flow
	UP	$\alpha = 1$, $\lvert W\rvert \lesssim 0.17$	$\lvert\mathrm{CFL}\rangle \otimes \overline{\lvert\mathrm{CFL}\rangle}$	gapless, flat $n(\mathbf{k})$
	UP	large $\lvert W\rvert$	Fermi liquid $\lvert\mathrm{FL}\rangle_{\mathrm{unpol.}}$	gapless, peaked $n(\mathbf{k})$
	UP	small $\lvert W\rvert$, neighbors VP	IVC	gapless, flat $n(\mathbf{k})$
$\nu = \frac{1}{2}$	VP	all α, $W = 0$	CFL $\lvert\mathrm{CFL}\rangle_{\mathrm{pol.}}$	flat $n(\mathbf{k})$
	VP	large $\lvert W\rvert$	Fermi liquid $\lvert\mathrm{FL}\rangle_{\mathrm{pol.}}$	peaked $n(\mathbf{k})$
	UP	large $\lvert W\rvert$	Fermi liquid $\lvert\mathrm{FL}\rangle_{\mathrm{unpol.}}$	peaked $n(\mathbf{k})$
	UP	$\alpha = 1$, $W = 0$	$\lvert\mathrm{CFL}\rangle \otimes \overline{\lvert\mathrm{CFL}\rangle}$	flat $n(\mathbf{k})$

Table 4.1: Numerical criteria used to identify the primary phases in (a), and candidate phases within the unresolved VP/UP regions in (b), for the lowest Landau level results.

The Hartree channel is nonzero only for $\mathbf{q} = 0$, while the exchange channel requires $\tau = \tau'$ and $\mathbf{k}' = \mathbf{k} + \mathbf{q}$, and picks up an overall minus sign from anticommuting the fermionic operators. This yields $\hat{H}_{HF} = \hat{H}_H + \hat{H}_X$, with

$$\hat{H}_H = \frac{1}{A} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\tau,\tau'} \tilde{U}_{\tau\tau'}(0)\, \Lambda_\tau(\mathbf{k}, 0)\Lambda_{\tau'}(\mathbf{k}', 0)\, \langle \hat{n}_{\mathbf{k},\tau} \rangle\, \hat{d}^\dagger_{\mathbf{k}',\tau'} \hat{d}_{\mathbf{k}',\tau'}, \tag{4.2}$$

$$\hat{H}_X = -\frac{1}{A} \sum_{\mathbf{k},\mathbf{q}} \sum_{\tau} \tilde{U}_{\tau\tau}(\mathbf{q})\, |\Lambda_\tau(\mathbf{k}, \mathbf{q})|^2\, \langle \hat{n}_{\mathbf{k}+\mathbf{q},\tau} \rangle\, \hat{d}^\dagger_{\mathbf{k},\tau} \hat{d}_{\mathbf{k},\tau}, \tag{4.3}$$

where $\hat{n}_{\mathbf{k},\tau} \equiv \hat{d}^\dagger_{\mathbf{k},\tau} \hat{d}_{\mathbf{k},\tau}$. Note that $\Lambda_\tau(\mathbf{k}, 0) = 1$ identically (the phase vanishes and $L_n(0) = 1$ for both $n = 0, 1$), so the form factor drops out of $\hat{H}_H$ entirely; this cancellation is used implicitly below.

4.2.1 Transition dependent on α at $W = 0$

At $W = 0$, the single-particle dispersion vanishes, so all single-particle states are exactly degenerate in energy regardless of momentum. There is thus no energy scale favoring any particular $\mathbf{k}$, which suggests a momentum-independent occupation, $\langle \hat{n}_{\mathbf{k},\tau} \rangle = \text{const.}$, as a natural ansatz, verified empirically in § 4.3.3. With $\sum_{\mathbf{k}} \langle \hat{n}_{\mathbf{k},\tau} \rangle = N_\phi \nu_\tau$, and using $\tilde{U}_{\tau\tau'}(0) = 2\pi d_{\mathrm{SC}}(1 + \alpha\tau\tau')$ (the $q \to 0$ limit of $\tilde{U}$, since $\tanh(q d_{\mathrm{SC}})/q \to d_{\mathrm{SC}}$), the Hartree–Fock energy becomes a function of the valley filling factors $\nu_\pm$ alone:

$$E_{HF}(\nu_+, \nu_-) = \frac{2\pi d_{\mathrm{SC}} N_\phi^2}{A} \left[(\nu_+ + \nu_-)^2 + \alpha(\nu_+ - \nu_-)^2 \right] - \frac{N_\phi(1+\alpha)}{A} \left(\nu_+^2 + \nu_-^2 \right) \sum_{\mathbf{q}} u_0(\mathbf{q}) e^{-q^2/2}, \tag{4.4}$$

where $u_0(\mathbf{q}) \equiv \frac{2\pi}{q(1 + l_{\mathrm{RK}} q)} \tanh(q d_{\mathrm{SC}})$ is the α-independent part of the interaction, and the Laguerre factor $L_n(q^2/2)$ has been set to 1 (we work in the lowest Landau level).

Comparing the energies of the fully valley-polarized state to the unpolarized state at fixed total filling $2\nu = \nu_+ + \nu_-$,

$$E(2\nu, 0) - E(\nu, \nu) = 2\nu^2 \frac{N_\phi}{A} \left[4\pi d_{\mathrm{SC}} N_\phi\, \alpha - (1 + \alpha) \sum_{\mathbf{q}} u_0(\mathbf{q}) e^{-q^2/2} \right]. \tag{4.5}$$

Defining $\mathcal{I} \equiv \frac{2\pi}{A} \sum_{\mathbf{q}} u_0(\mathbf{q}) e^{-q^2/2}$ and using $1/N_\phi = 2\pi/A$, the valley-polarized (VP) state is energetically favored, $E(2\nu, 0) - E(\nu, \nu) < 0$, when

$$4\pi d_{\mathrm{SC}}\, \alpha - (1 + \alpha)\, \mathcal{I} < 0 \quad \Longleftrightarrow \quad \alpha\, (4\pi d_{\mathrm{SC}} - \mathcal{I}) < \mathcal{I}. \tag{4.6}$$

Dividing by $(4\pi d_{\mathrm{SC}} - \mathcal{I})$ and tracking the sign of the denominator gives the phase boundary

$$\alpha < \frac{1}{\dfrac{4\pi d_{\mathrm{SC}}}{\mathcal{I}} - 1} \quad \text{for} \quad 4\pi d_{\mathrm{SC}} > \mathcal{I}, \tag{4.7}$$

$$\alpha > \frac{1}{\dfrac{4\pi d_{\mathrm{SC}}}{\mathcal{I}} - 1} \quad \text{for} \quad 4\pi d_{\mathrm{SC}} < \mathcal{I}. \tag{4.8}$$

In the continuum limit, $\frac{1}{A} \sum_{\mathbf{q}} \to \int \frac{d^2 q}{(2\pi)^2} = \frac{1}{2\pi} \int_0^\infty q\, dq$, so

$$\mathcal{I}(d_{\mathrm{SC}}, l_{\mathrm{RK}}) = 2\pi \int_0^\infty dq\, \frac{\tanh(q d_{\mathrm{SC}})}{1 + l_{\mathrm{RK}} q}\, e^{-q^2/2}. \tag{4.9}$$

4.2.2 Transition dependent on W

The depolarizing transition driven by increasing W was treated with a Stoner-like criterion in Ref. [39], assuming a dispersion of the form $\varepsilon(\mathbf{k}) \propto |\mathbf{k}|^m$, $m \in \mathbb{N}$; this is not a good approximation to the harmonic dispersion used here, and combined with the general tendency of Hartree-Fock-level criteria to overestimate the stability of ordered phases, gives an order-of-magnitude error in the critical W.

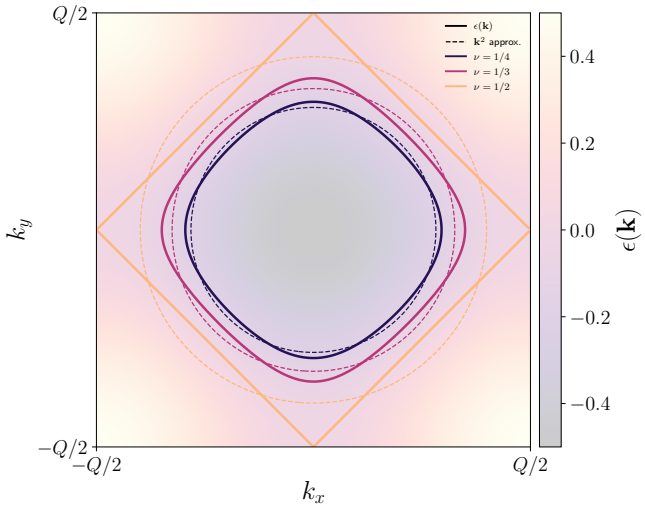


Figure 4.1: Dispersion $\varepsilon(\mathbf{k})$ at $W = 1.0$, with the Fermi surfaces at fillings $\nu = 1/4, 1/3, 1/2$ shown both for the actual (cosine) dispersion and for a parabolic approximation.

As shown in Fig. 4.1, the Fermi surface (and even more so the density of states) of the actual dispersion differs significantly from that of the parabolic approximation, especially near the Van Hove singularity approached at $\nu = \frac{1}{2}$. We therefore relax this assumption: a similar criterion is derived below in which the Fermi surface is still taken to be circular for simplicity, but the density of states is obtained by numerically integrating the actual dispersion rather than assumed in a closed form.

Here W is assumed to be large enough that the occupation follows a Fermi–Dirac step, $n_\tau(\mathbf{k}) = \theta(k_{F\tau} - |\mathbf{k}|)$ with $k_{F\tau} = \sqrt{2\nu_\tau}$.

Hartree energy ($\alpha = 0$). Since $\tilde{U}_{\tau\tau'}(0) = 2\pi d_{\mathrm{SC}}$ is independent of valley at $\alpha = 0$,

$$\langle \hat{H}_H \rangle = \frac{2\pi d_{\mathrm{SC}}}{A} \sum_{\mathbf{k},\mathbf{k}'} \sum_{\tau,\tau'} \langle n_{\mathbf{k},\tau} \rangle \langle n_{\mathbf{k}',\tau'} \rangle = \frac{2\pi d_{\mathrm{SC}} N_\phi^2}{A} (\nu_+ + \nu_-)^2, \tag{4.10}$$

independent of the valley polarization.

Exchange energy. At $\alpha = 0$, $\tilde{U}_{\tau\tau}(\mathbf{q}) = u_0(\mathbf{q})$, giving

$$\langle \hat{H}_X \rangle = -\frac{1}{A} \sum_{\mathbf{q}} u_0(\mathbf{q}) e^{-q^2/2} \sum_{\tau,\mathbf{k}} n_\tau(\mathbf{k}+\mathbf{q})\, n_\tau(\mathbf{k}). \tag{4.11}$$

Converting sums to integrals ($\sum_{\mathbf{k}} \to \frac{A}{(2\pi)^2} \int d^2k$, and similarly for $\mathbf{q}$),

$$E_X = -\frac{A}{(2\pi)^4} \int d^2q\, u_0(\mathbf{q}) e^{-q^2/2} \sum_\tau \int d^2k\, \theta(k_{F\tau} - |\mathbf{k}+\mathbf{q}|) \theta(k_{F\tau} - |\mathbf{k}|). \tag{4.12}$$

The inner integral is exactly the overlap area of two disks of radius $k_{F\tau}$ with centers separated by q,

$$F(k_{F\tau}, q) = 2k_{F\tau}^2 \arccos\left(\frac{q}{2k_{F\tau}}\right) - \frac{q}{2}\sqrt{4k_{F\tau}^2 - q^2}, \tag{4.13}$$

so that

$$E_X = -\frac{A}{(2\pi)^2}\int \frac{d^2q}{(2\pi)^2}\, u_0(\mathbf{q}) e^{-q^2/2} \sum_\tau F(k_{F\tau}, q). \tag{4.14}$$

Expansion around the unpolarized point. Write $\nu_\tau = \nu + \tau\delta$, so that δ is the polarization ($\delta = 0$ corresponds to the unpolarized state), and $k_{F\tau} = \sqrt{2\nu_\tau}$, approximating the Fermi surface as a circle. Expanding F to second order in δ and integrating over angle,

$$\left.\frac{d^2E_X}{d\delta^2}\right|_{\delta=0} = -\frac{4A}{\nu}\int_0^{\sqrt{8\nu}} \frac{q\tanh(qd_{\mathrm{SC}})\, e^{-q^2/2}}{(2\pi)^2(1 + l_{\mathrm{RK}}q)\sqrt{8\nu - q^2}}\, dq \;\equiv\; -\frac{4A}{\nu}\mathcal{I}_2(\nu, d_{\mathrm{SC}}). \tag{4.15}$$

The kinetic-energy curvature follows directly from the dispersion, obtained numerically via the density of states $D(\nu) = d\mu/d\nu$ computed from $\varepsilon_\tau(\mathbf{k})$:

$$\left.\frac{d^2E_K}{d\delta^2}\right|_{\delta=0} = 2AW\frac{d\mu}{d\nu} = \frac{AW}{\pi}D(\nu). \tag{4.16}$$

The transition occurs where the curvature of the unpolarized state changes sign, i.e. where the kinetic cost of polarizing exactly cancels the exchange gain:

$$\left.\frac{d^2E_K}{d\delta^2}\right|_{\delta=0} + \left.\frac{d^2E_X}{d\delta^2}\right|_{\delta=0} = 0 \quad\Longrightarrow\quad \frac{AW_c}{\pi}D(\nu) = \frac{4A}{\nu}\mathcal{I}_2(\nu, d_{\mathrm{SC}}), \tag{4.17}$$

giving

$$W_c(\nu, d_{\mathrm{SC}}) = \frac{4\pi\,\mathcal{I}_2(\nu, d_{\mathrm{SC}})}{\nu\, D(\nu)}. \tag{4.18}$$

This result extends straightforwardly to nonzero α via the same procedure, giving

$$W_c(\nu, d_{\mathrm{SC}}, \alpha) = \frac{4\pi\,(1+\alpha)\mathcal{I}_2(\nu, d_{\mathrm{SC}})}{\nu\, D(\nu)} - \frac{4\,\alpha d_{\mathrm{SC}}}{D(\nu)}. \tag{4.19}$$

4.2.3 Comparison with the numerical mean-field estimates

We evaluate the analytic estimates of Secs. 4.2.1–4.2.2 numerically at the default parameters $d_{\mathrm{SC}} = 6a$, $l_{\mathrm{RK}} = 0$, using $D(\nu)$ obtained from the actual cosine dispersion (Fig. 4.1) rather than the parabolic approximation, and compare against the ED phase diagrams below.

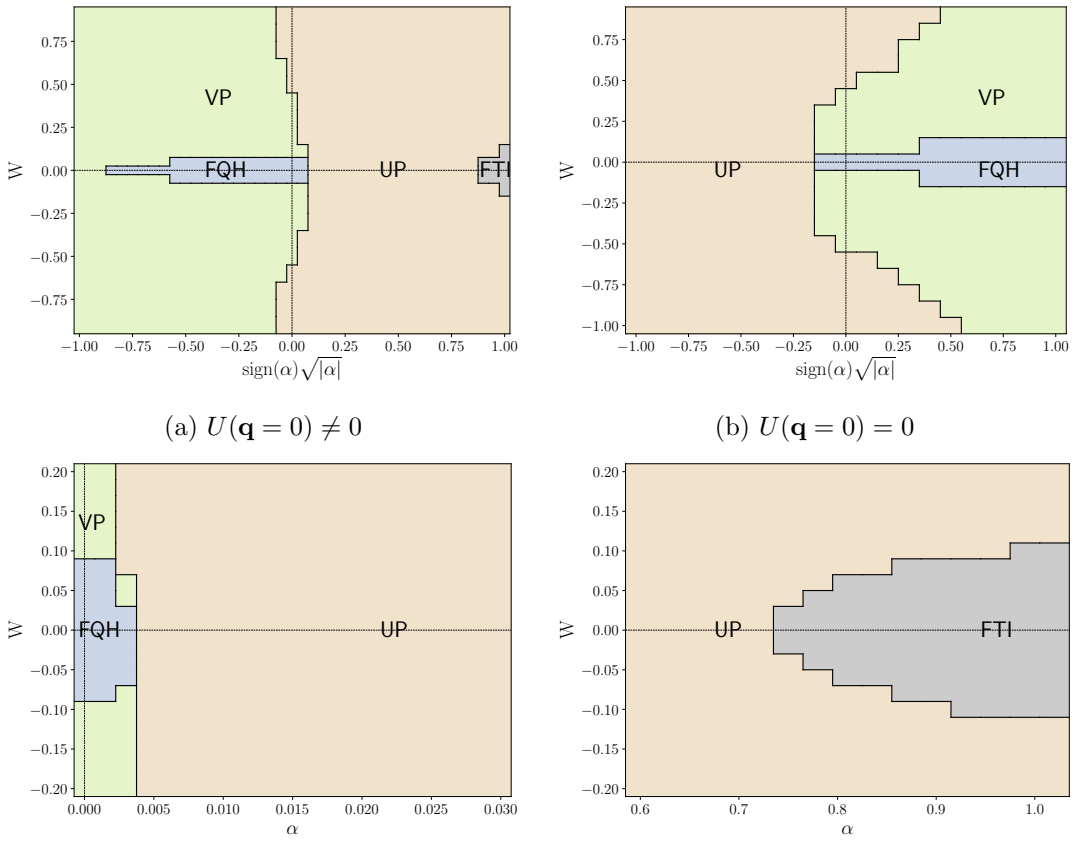
At $W = 0$, Eqs. (4.7)–(4.8) give $\alpha_c \approx 0.042$ at $\nu = \frac{1}{3}$ per valley, compared to the ED value $\alpha_c \approx 0.004$ extracted from Fig. 4.2a. The mean-field estimate thus overestimates α_c by roughly an order of magnitude, but correctly predicts the direction of the polarized phase ($\alpha < \alpha_c$ in the $U(\mathbf{q} = 0) \neq 0$ case), proximity to the valley-isotropic point and, more importantly, its trend: scanning $\alpha_c(d_{\mathrm{SC}})$ and $\alpha_c(l_{\mathrm{RK}})$ analytically gives a monotonic decrease with both parameters, matching the monotonic narrowing of the α-stable FQH region with increasing d_{SC} and l_{RK} observed in Figs. 4.9a and 4.9b.

For the W-driven depolarization transition, the mean-field estimate at $\nu = \frac{1}{3}$, $\alpha = 0$ gives $W_c \approx 0.84$, compared to the ED result of $W \sim 0.5$ read off from Fig. 4.2a. As with α_c, the analytic estimate is systematically too large but of the correct order of magnitude, consistent with the general tendency of Hartree-Fock to overestimate the stability of ordered phases by neglecting quantum fluctuations.

Overall, the analytic mean-field treatment reproduces the correct qualitative structure of the phase diagram — the existence and approximate location of both the α- and W-driven depolarization transitions, and their correct monotonic dependence on the screening parameters d_{SC} and l_{RK} — while systematically overestimating both α_c and W_c.

4.3 Phase diagram for $\nu = \frac{2}{3}$

At this filling, an FQH state is expected at $W = 0$ in the valley-polarized region, and an FTI state at $\alpha = 1$, $W = 0$. We now present the numerical phase diagram, addressing how resilient these topological phases are against nonzero $|W|$ and non-ideal α, and examine phase transitions found in the system, and the dependence of stability of topological phases on the screening parameters d_{SC} and l_{RK}.



(a) $U(\mathbf{q} = 0) \neq 0$

(b) $U(\mathbf{q} = 0) = 0$

(c) Zoom-in on the FQH ($U(\mathbf{q} = 0) \neq 0$)

(d) Zoom-in on the FTI ($U(\mathbf{q} = 0) \neq 0$)

Figure 4.2: Phase diagrams of the $\nu = \frac{2}{3}$ filling. VP: gapless valley-polarized state. UP: general unpolarized state. FQH: 3-fold degenerate valley-polarized fractional quantum Hall state. FTI: 9-fold degenerate valley-unpolarized fractional topological insulator. See Fig. 6.1 for a schematic identification of the UP regions.

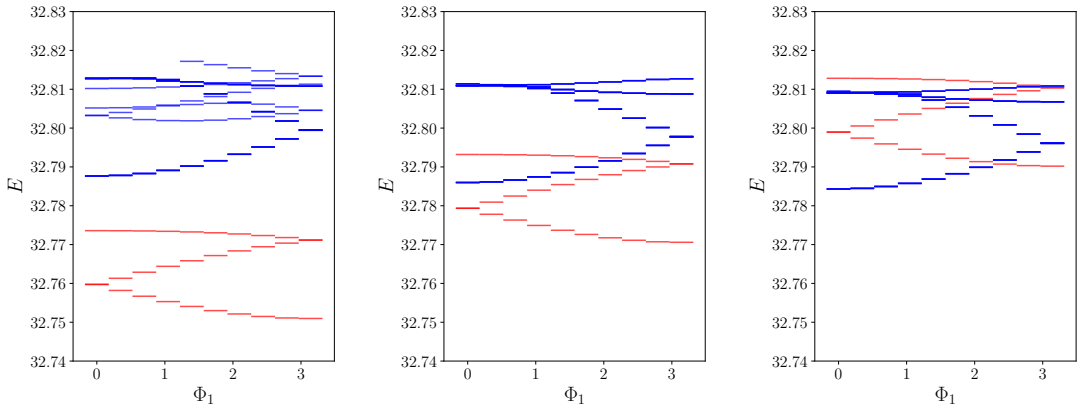
Figure 4.2 presents the phase diagram at $\nu = \frac{2}{3}$. The only significant gaps found in the system are Δ_3, Δ_6 (discussed below) and Δ_9. In the $U(\mathbf{q} = 0) \neq 0$ case (Fig. 4.2a), full valley polarization is observed at low $|W|$ and low α, with $\alpha_c \approx 0.004$, while the unpolarized state prevails at larger α. At $\alpha = 1$, down to $\alpha \approx 0.72$, the unpolarized state shows a clear 9-fold degeneracy, consistent with the FTI state (each FQH copy contributing a 3-fold degeneracy). The valley-polarized region itself splits into two parts. At low $|W|$ the state has a topological 3-fold degeneracy, consistent with a $\nu = \frac{2}{3}$ FQH state in a single valley. The topological nature of FQH and FTI phases is confirmed below via spectral flow. Both of them survive a nonzero bandwidth, up to $|W| \approx 0.09$ for the FQH state and $|W| \approx 0.1$ for the FTI state. The similarity of these two values is consistent with the ideal FQH and FTI states incurring the same kinetic energy penalty from W. At intermediate $|W|$, the FQH state transitions into a gapless valley-polarized state (denoted as VP) — driven by the kinetic-energy cost of the flat momentum occupation characteristic of the FQH state — and, at an α-dependent W_c on the order of ± 0.5,

into a valley-unpolarized state. The FTI state undergoes an analogous transition into a gapless state at increasing W.

The phase diagram is approximately symmetric about the $W = 0$ axis, as suspected from the analysis in §2.3. The symmetry being only approximate is a finite size effect, in particular related to the fact that the sampling of the Brillouin zone is not invariant under the $\pm\mathbf{Q_0}$ shift. At $\alpha \to -1$, the FQH state is not detected, since valley polarization combined with the vanishing intra-valley interaction renders the system effectively noninteracting, closing the spectral gap.

A different picture emerges for the $U(\mathbf{q} = 0) = 0$ phase diagram (Fig. 4.2b), where the valley-polarized region instead occupies large α, and the unpolarized — small α. The same VP and FQH phases appear, with a similar W-dependent transition between them, but because the polarized phase now extends across the entire $\alpha > 0$ region, no FTI state is found. At low α, only a non-polarized gapless state is found — it is not investigated further.

4.3.1 Phase transitions at $W \approx 0$ and variable α



(a) $\alpha = 0.0024$ — FQH (b) $\alpha = 0.003$ — gapless (c) $\alpha = 0.0036$ — gapless

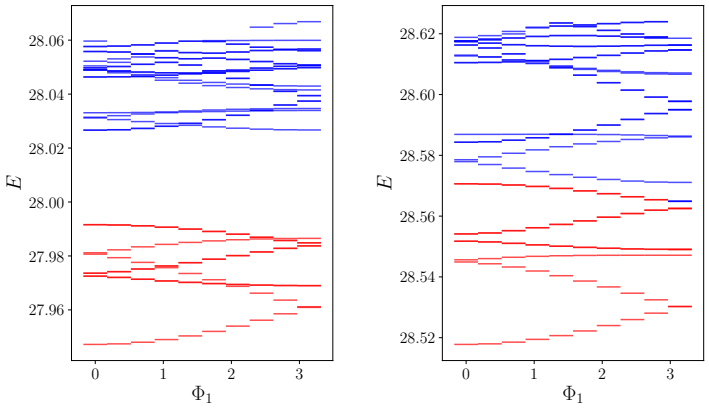
Figure 4.3: Flux insertion plots of the energy spectrum near the valley-isotropic limit $\alpha \approx 0$, at $W = 0.05$ and $\nu = \frac{2}{3}$, showing the transition from the FQH state to a gapless state. Ground-state manifold shown in red.

The nature of the phase transition out of the FQH state is illustrated in Figure 4.3, which shows the energy spectrum under flux insertion at $W = 0.05$ for different α values, across the transition out of the FQH phase. Interestingly, the 3-fold degenerate ground-state manifold survives the transition; the only change is the relative energy of the manifold with respect to the gapless continuum above it.

Figure 4.4 shows the analogous phase winding at $W = 0.05$ with α decreasing from the FTI-like state. As before, the ground-state manifold moves relative to the excited-state continuum until the 9-fold degenerate gap closes. However, the relative energy splitting between the lowest 3 states of the ground-state manifold and the remaining 6 grows on approach to the transition, eventually opening a gap between them: a nonzero 3-fold gap in the otherwise unpolarized state is indeed found in a small region of the phase diagram. As described below (§4.3.3), this phase is not topological.

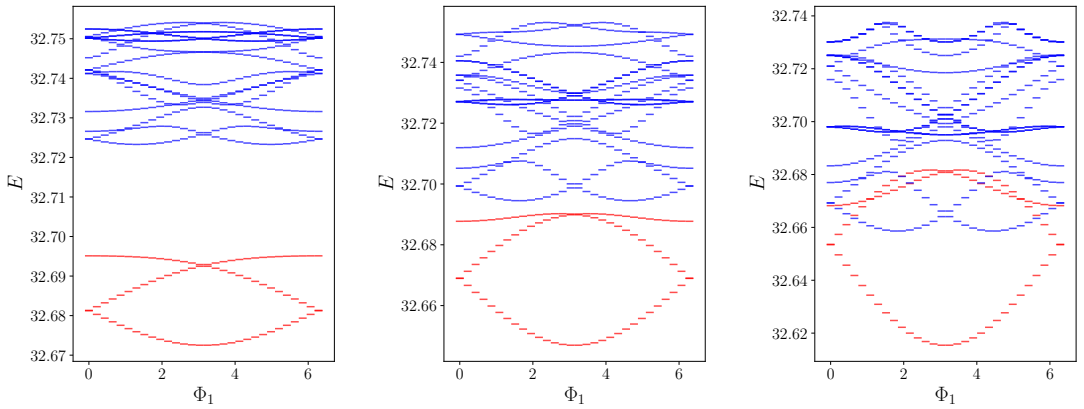
4.3.2 Spectral flow & phase transitions with variable W

Figures 4.5 and 4.6 show the energy spectrum for the FQH ($\alpha = 0.0$) and FTI ($\alpha = 1$) phases respectively, under flux insertion. At low W (Figs. 4.5a and 4.6a), spectral flow of the ground-state manifold is directly observed: the spectrum at $\Phi_1 = 0$ and $\Phi_1 = 2\pi$ is identical, but the



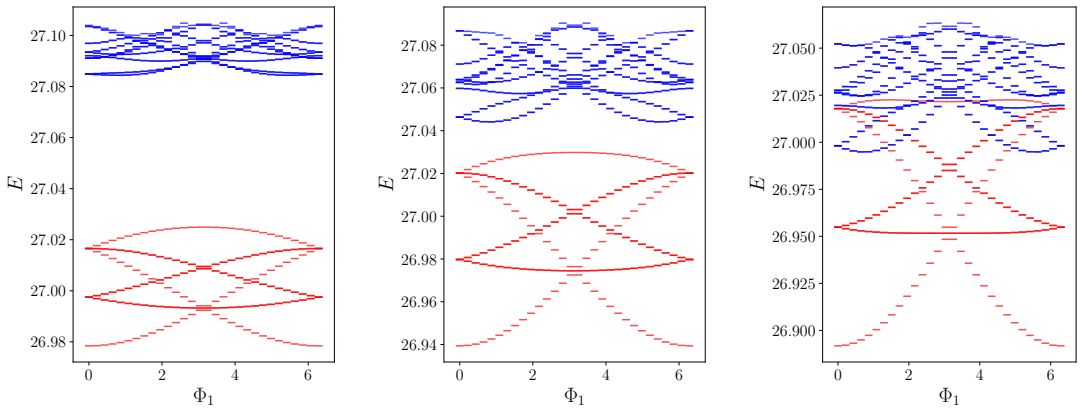
(a) $\alpha = 0.85$ — FTI state (b) $\alpha = 0.76$ — gapless state

Figure 4.4: Phase winding of the energy spectrum near the decoupled limit $\alpha \approx 1$, at $W = 0.05$ and $\nu = \frac{2}{3}$, showing the transition from the FTI state to a gapless state. Ground-state manifold shown in red.



(a) $W = 0.05$ (b) $W = 0.1$ (c) $W = 0.15$

Figure 4.5: Phase winding of the energy spectrum at $\alpha = 0$ and $\nu = \frac{2}{3}$, showing the transition from the FQH state to a gapless state with increasing W. Ground-state manifold shown in red.

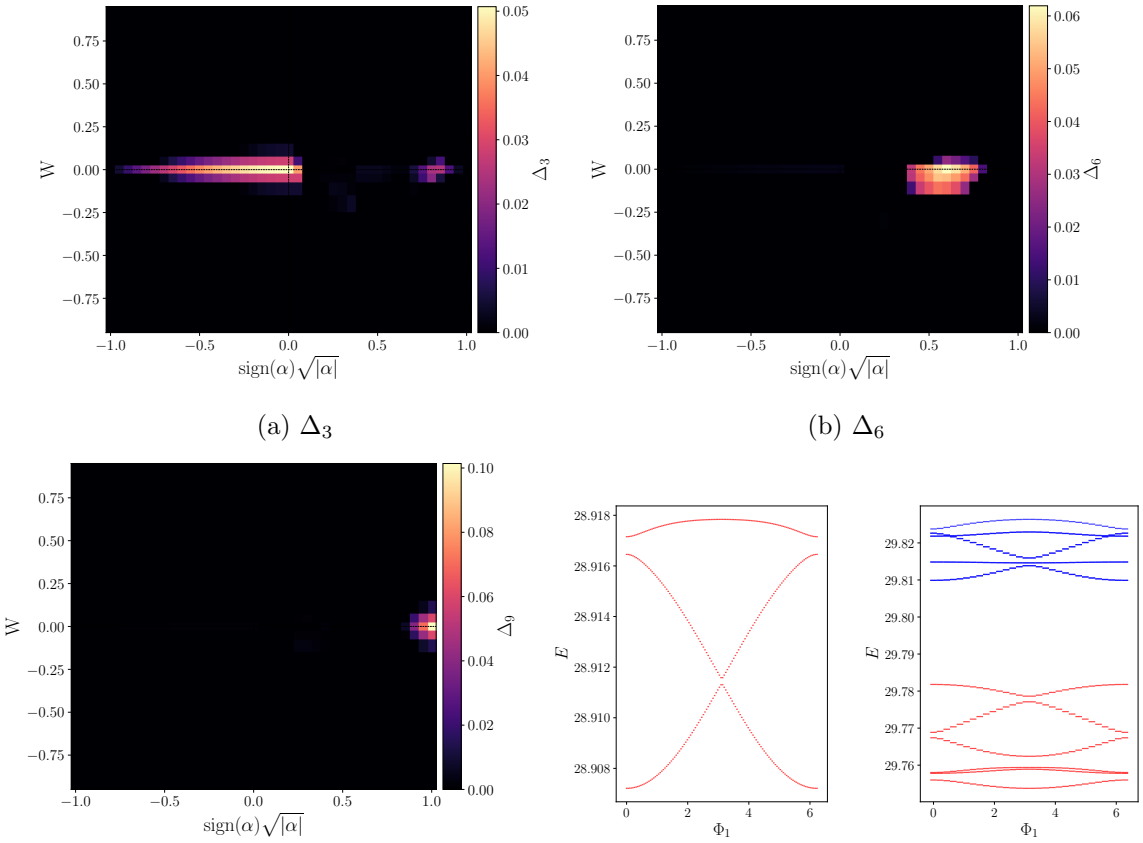


(a) $W = 0.05$ (b) $W = 0.1$ (c) $W = 0.15$

Figure 4.6: Phase winding of the energy spectrum at $\alpha = 1$ and $\nu = \frac{2}{3}$, showing the transition from the FTI state to a gapless state with increasing W. Ground-state manifold shown in red.

states within the manifold are cyclically permuted, returning to their original positions only after $\Phi_1 = 6\pi$. This further confirms the topological nature of both phases. The remaining panels of each figure visualize the transition from the topological to the gapless phase as the increasing bandwidth closes the spectral gap.

4.3.3 Nature of the intermediate–α and high-$|W|$ phases



(a) Δ_3 (b) Δ_6

(c) Δ_9 (d) $\alpha = 0.7$, $W = 0.025$ (e) $\alpha = 0.56$, $W = 0.025$

Figure 4.7: (a-c) Spectral gaps in the system as a function of α and W, at $\nu = \frac{2}{3}$. (d-e) Flux insertion of the 3-fold and 6-fold degenerate states, respectively.

Figure 4.7 shows the size of the Δ_3, Δ_6, and Δ_9 gaps across the phase diagram. It confirms directly that the FQH and FTI states are most stable at $W = 0$, where the corresponding gap is largest. Notably, non-zero Δ_3 and Δ_6 gaps are also found in the unpolarized phase at intermediate α. However, as shown in Fig. 4.7e and 4.7d, both the 3-fold and the 6-fold degenerate states do not exhibit spectral flow — the energy levels do not cross or permute — and are therefore not topological. It could correspond to a charge-density-wave state or be an artifact of a finite system size. Because of their uncertain nature and small extent, they were not investigated further and are included under the umbrella "UP" region of the phase diagram.

To further examine the UP region and the assumptions underlying the mean-field treatment of Sec. 4.2, the momentum-occupation function $n(\mathbf{k})$ is also calculated.

Figure 4.8 shows that the occupation is essentially flat at $W \approx 0$, while its standard deviation grows large at large $|W|$, confirming the expected Fermi-liquid character of the large-$|W|$ phases. The gapless VP phase likewise shows a non-flat occupation, consistent with Fermi-liquid behavior there. Two additional regions of low standard deviation stand out, surviving up to a discontinuity at relatively large $|W|$. The lower of these regions may correspond to an IVC

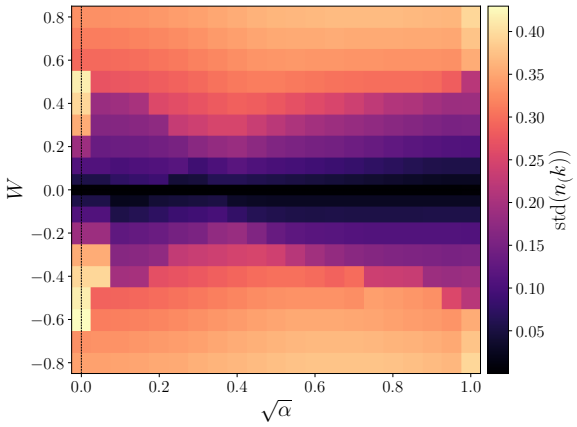
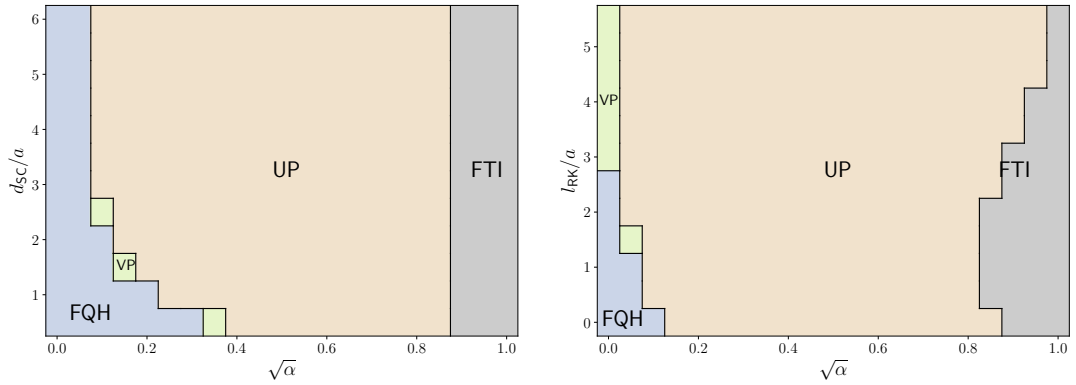


Figure 4.8: Standard deviation of $n(\mathbf{k})$ over the Brillouin zone as a function of α and W, at $\nu = \frac{2}{3}$.

phase undergoing a Stoner-like transition into a FL state analogous to the case of the depolarizing transition from the VP state, but with a different order parameter. This would agree with Ref. [39], where the VP state is expected to prevail at $\alpha < 0$ and the IVC at $\alpha > 0$, both with phase transitions of the same nature. A schematic summary including these candidate phases is given in Fig. 6.1. The larger–α flat occupation region is not studied in detail here.

4.3.4 Screening of the interaction



(a) α vs. d_{SC}/a, at $l_{\mathrm{RK}} = 0$

(b) α vs. l_{RK}/a, at $d_{\mathrm{SC}} = 2a$

Figure 4.9: Phase diagrams of the $\nu = \frac{2}{3}$ filling at $W = 0$ with respect to the screening parameters.

The stability of the topological phases was tested against the two screening parameters d_{SC} and l_{RK}. Figure 4.9a shows the extent of the phases at $W = 0$ as a function of α and the screening length d_{SC}. Two conclusions follow. First, the α-stability of the FQH phase decreases monotonically with increasing d_{SC}, in agreement with the analytic mean-field estimate, where increasing d_{SC} shifts the α_c phase boundary in the same direction. Second, no corresponding change in the stability of the FTI phase is visible.

Figure 4.9b shows the analogous dependence on l_{RK} at $W = 0$. Here the stability of both topological phases varies: the FQH phase again becomes less stable with increasing l_{RK}, while the FTI phase instead shows a non-monotonic dependence, with a maximum of stability at finite l_{RK}.

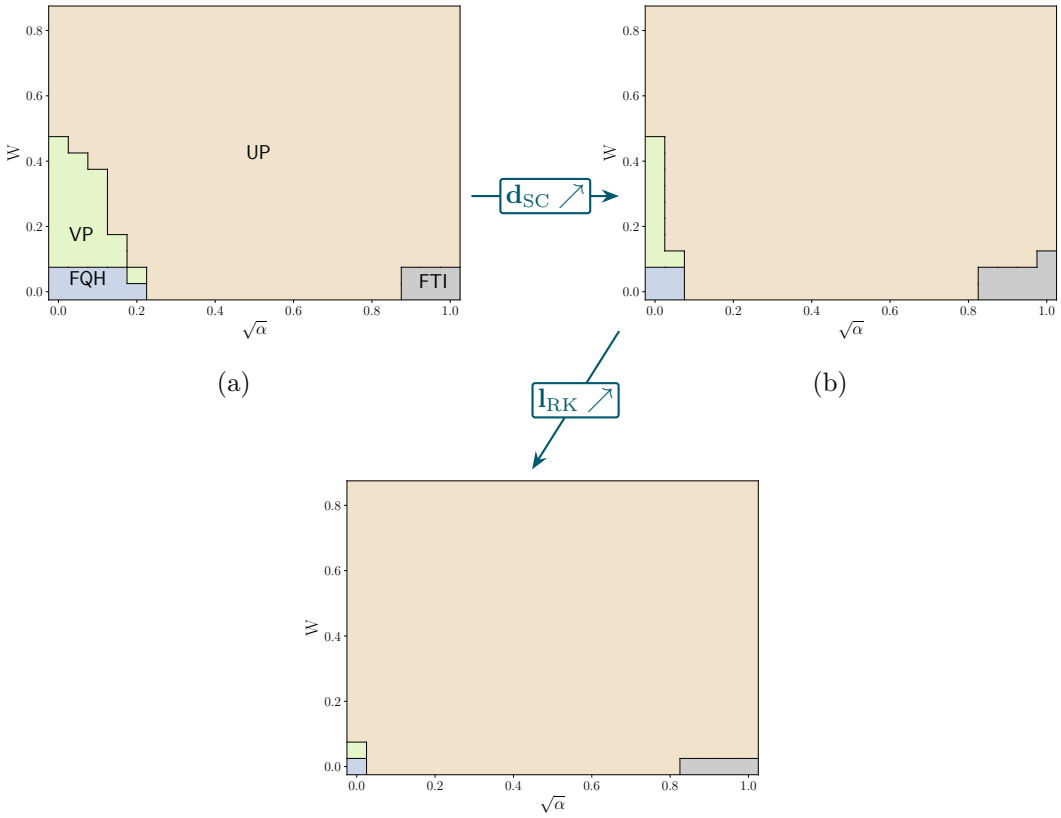
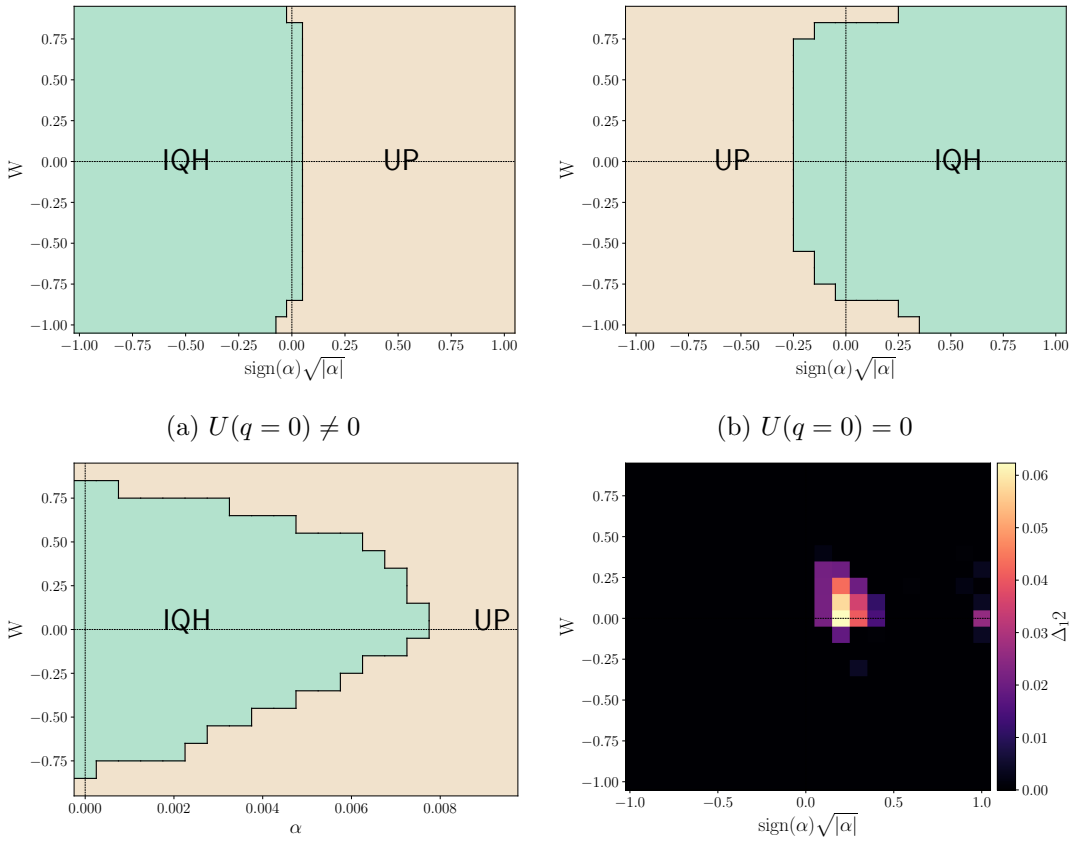


Figure 4.10: Phase diagrams of the $\nu = \frac{2}{3}$ filling in the α–W plane with different screening d_{SC} and l_{RK}. Arrows indicate the direction of change of the parameters. (a) $d_{\mathrm{SC}} = 1a$, $l_{\mathrm{RK}} = 0$. (b) $d_{\mathrm{SC}} = 6a$, $l_{\mathrm{RK}} = 0$. (c) $d_{\mathrm{SC}} = 6a$, $l_{\mathrm{RK}} = 2a$.

Figure 4.10 shows the full α–W phase diagram at different screening parameters, allowing the W-stability to be examined alongside the α-stability. The main conclusion is that the W-stability is essentially independent of d_{SC}, but decreases significantly with increasing l_{RK}. This is particularly relevant for the FTI phase, where seemingly to gain α-stability by tuning the screening, one must sacrifice the W-stability.

4.4 Phase diagram for $\nu = 1$

While fractional topological states are not expected at $\nu = 1$ in the LLL, this filling has been studied in the literature and serves as a useful benchmark for the method, as well as a reference point for $n = 1$ LL calculations in Chapter 5 [12, 23].



(a) $U(q = 0) \neq 0$

(b) $U(q = 0) = 0$

(c) Zoom-in on the IQH ($U(q = 0) \neq 0$)

(d) Δ_{12} gap, $U(q = 0) \neq 0$

Figure 4.11: Phase diagrams of the $\nu = 1$ filling. UP: general unpolarized state. IQH: valley-polarized integer quantum Hall state. See Fig. 6.1 for a schematic identification of the UP regions.

Figure 4.11 shows the resulting phase diagram. With $U(\mathbf{q} = 0) \neq 0$, similarly to the $\nu = \frac{2}{3}$, a valley-polarized region is found at low α, giving way to a gapless non-polarized phase at larger α or larger $|W|$. Due to integer filling, the valley-polarized phase is the IQH state at $\nu = 1 + 0$ and with $\Delta_1 > 0$. The resulting phase boundary is consistent with past results [23] and with the numerical mean-field estimate, within the same order of magnitude.

The nature of the high–α gapless state can again be discussed in light of the literature and the momentum-occupation function. In the $\alpha = 1$, $W = 0$ limit, the $\nu = \frac{1}{2} + \frac{1}{2}$ state is known to consist of decoupled composite Fermi liquids, $|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$, while at large bandwidth $|W|$, where kinetic energy dominates, it becomes a pair of Fermi liquids, $|\mathrm{FL}\rangle \otimes \overline{|\mathrm{FL}\rangle}$. The phase

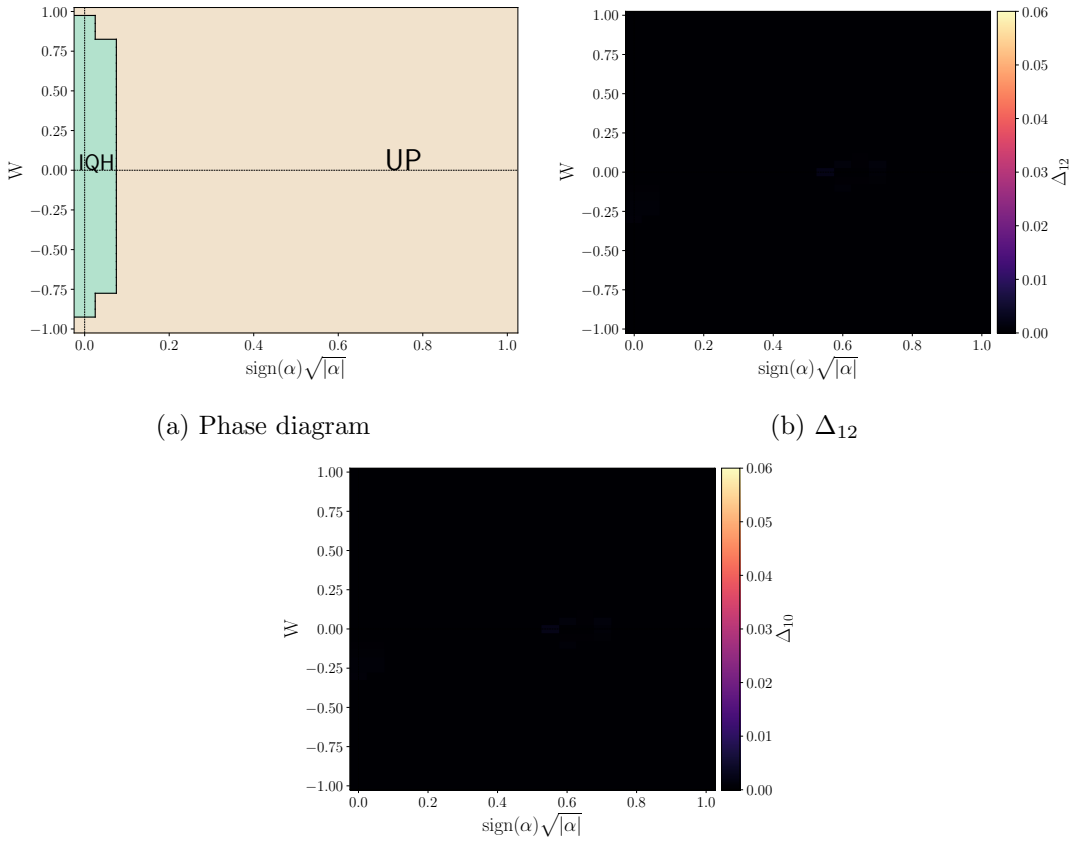
transition and boundary between these two states, and the nature of the intermediate states once $\alpha \neq 1$ cannot be resolved with the limited extent of the simulations performed here.

With $U(\mathbf{q} = 0) = 0$, the phase diagram shown in Figure 4.11b is again inverted. While containing the same phases (IQH, VP and UP) and the same W dependence, now the ground state is unpolarized at low α while the IQH is prevalent at high α.

4.4.1 Intermediate–α and high-$|W|$ phases

Finite size effects

Returning to the $U(\mathbf{q} = 0) \neq 0$ case, at intermediate α, a $\Delta_{12} \neq 0$ gap is found (Fig. 4.11d), similarly to the $\nu = \frac{2}{3}$ system. Such degeneracy does not correspond to any topologically nontrivial state and phase winding is not present. Since the degeneracy is equal to the number of unit cells, it suggests complete translational symmetry breaking in a form of phase separation. It is important to note that the gap is nonzero only for $W \geq 0$, which suggests that the state might stem from finite size effects. An IVC state was reported in this region of the phase diagram by Ref. [23], with the peak of the IVC order parameter Δ_q at $q = \mathbf{Q_0} = (\frac{Q}{2}, \frac{Q}{2})$. The default 3×4 BZ sampling studied here does not contain this point, which might result in favoring other competing phases. To clarify the nature of the state an additional calculation is performed at the same filling, with $N_\phi = 10$, in a tilted lattice of Figure 5.1a. This lattice, on the other hand, contains the $\mathbf{Q_0}$ point. The sampling is invariant under the $\mathbf{Q_0}$ shift, and is also more isotropic with $\frac{|T_1|}{|T_2|}$ closer to 1.



(a) Phase diagram

(b) Δ_{12}

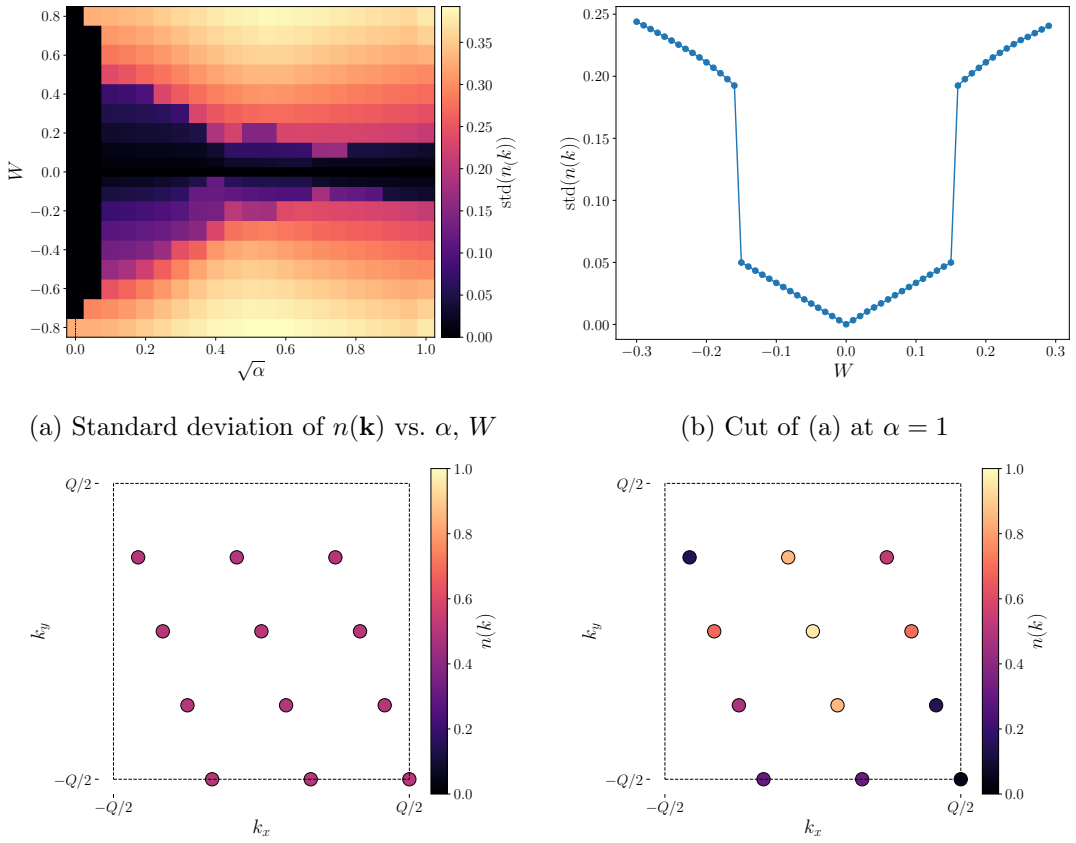
(c) Δ_{10}

Figure 4.12: Phase diagram and gaps in a $\nu = 1$ system with tilted boundary conditions of Fig. 5.1a.

The results are shown in Figure 4.12. First, in this system both the Δ_{12} gap — as a

comparison with the 3×4 system — and the Δ_{10} gap — corresponding now to the number of unit cells in the system — are strictly 0, confirming that the gapped phase is favored only due to finite size effects or anisotropy of the cluster. Second, in Figure 4.12a the phase diagram is perfectly symmetric with respect to $W = 0$, confirming the prediction of §2.3 to be exact when the BZ sampling is invariant under the $\mathbf{Q_0}$ shift.

Momentum-occupation function



(a) Standard deviation of $n(\mathbf{k})$ vs. α, W

(b) Cut of (a) at $\alpha = 1$

(c) $n(\mathbf{k})$ at $W = 0.0$, $\alpha = 1.0$
$|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$

(d) $n(\mathbf{k})$ at $W = 0.5$, $\alpha = 1.0$
$|\mathrm{FL}\rangle \otimes \overline{|\mathrm{FL}\rangle}$

Figure 4.13: The momentum-space occupation $n(\mathbf{k})$ for the $\nu = 1$ filling, shown for an arbitrary valley.

Figure 4.13 shows the standard deviation of $n(\mathbf{k})$ over the Brillouin zone. A large region of flat occupation is again observed at $\sqrt{\alpha} \in [0.05, 0.4]$, which, by comparison with the literature, may correspond to the IVC state, with a discontinuous transition to the Fermi liquid at higher $|W|$: while Kwan et al. [23] confirm the IVC phase, the transition itself is not reported there — IVC being also found at higher W. The same discontinuity of $n(\mathbf{k})$ is found both at $W > 0$ and at $W < 0$ in both the default and the tilted cluster calculations, confirming that it is not caused by the phase-separated state found only in one of these four cases.

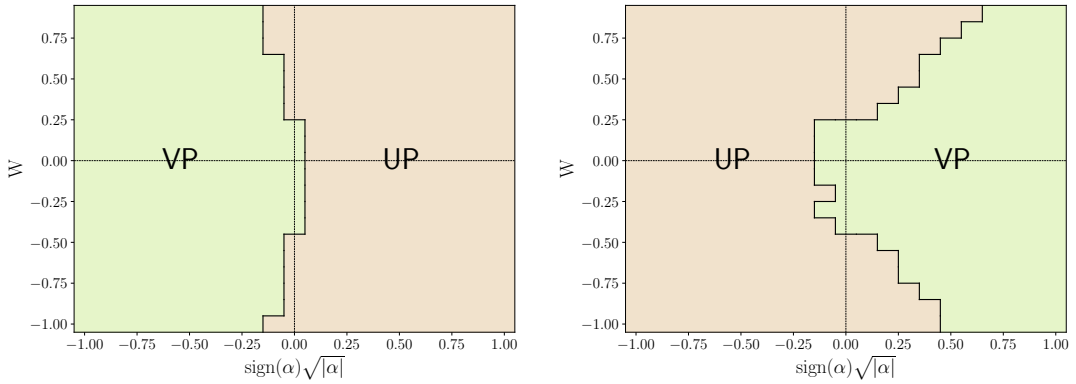
The IVC phase reported in Ref. [23] is a gapped Chern-texture insulator, while the state found here is consistently gapless. This is not incompatible with their result, since their Hartree–Fock gap is a charge gap, whereas the gap calculated here is a neutral gap, which is expected to be gapless in the IVC phase due to its Goldstone mode.

Additionally, a clear transition between $|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$ and $|\mathrm{FL}\rangle \otimes \overline{|\mathrm{FL}\rangle}$ is seen at $\alpha = 1$, with a jump in the standard deviation of $n(\mathbf{k})$ at $W \approx 0.17$ (Fig. 4.13b); an analogous transition for

a single Landau level was studied by Barkeshli and McGreevy [2]. The momentum-occupation function is shown explicitly on either side of this transition: it is flat at low $|W|$ (Fig. 4.13c), and shows a clear Fermi surface at high $|W|$ (Fig. 4.13d).

4.5 Phase diagram for $\nu = \frac{1}{2}$

For completeness, the $\nu = \frac{1}{2}$ filling was studied in an analogous way. No gapped topological order is expected here in the lowest Landau level, but the valley polarization is expected to behave qualitatively similarly to the other fillings considered above.



(a) $U(\mathbf{q} = 0) \neq 0$ (b) $U(\mathbf{q} = 0) = 0$

Figure 4.14: Phase diagrams of the $\nu = \frac{1}{2}$ filling. VP: gapless valley-polarized state. UP: general unpolarized state. See Fig. 6.1 for a schematic identification of the VP/UP regions.

Figure 4.14 shows the resulting phase diagram. With $U(\mathbf{q} = 0) \neq 0$, two gapless phases are present: a VP phase at low α or low $|W|$, and a non-polarized phase otherwise. Following the same reasoning as for $\nu = 1$, the $\alpha = 1$, $W = 0$ limit is expected to host a $|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$ state at $\nu = \frac{1}{4} + \frac{1}{4}$, while the gapless phase at $\alpha = 1$ and high $|W|$ is expected to be $|\mathrm{FL}\rangle \otimes \overline{|\mathrm{FL}\rangle}$; the intermediate parameter regime is harder to characterize unambiguously.

Unlike at the other fillings, the VP state here is similar in nature to the unpolarized state. At $W = 0$, the fully valley-polarized system is a $\mathrm{CFL}_{\mathrm{pol.}}$ for all relevant α, since with only one valley occupied the inter-valley coupling plays no role; a Stoner-like transition from the valley-polarized to the unpolarized state is again found at large $|W|$. Once again, the phase diagram is approximately symmetric around the $W = 0$ axis.

For the $U(\mathbf{q} = 0) = 0$ (Figure 4.14b) the same phases are again detected but with the unpolarized region at low α instead of large α. The same Stoner-like transition is found at high $|W|$.

4.5.1 Momentum-occupation function

As a further diagnostic for the phases in $U(\mathbf{q} = 0) \neq 0$ case, Fig. 4.15 shows the standard deviation of $n(\mathbf{k})$ over the Brillouin zone: the occupation is flat at low $|W|$ and develops a clear Fermi surface at high $|W|$. Even though the valley-polarized region is gapless throughout, a clear distinction remains within it between the flat occupation characteristic of a CFL at low $|W|$ and the peaked Fermi surface of an FL at intermediate $|W|$. While not studied in further detail here, a phase transition analogous to the one found at $\nu = 1$ is suspected between these states as well.

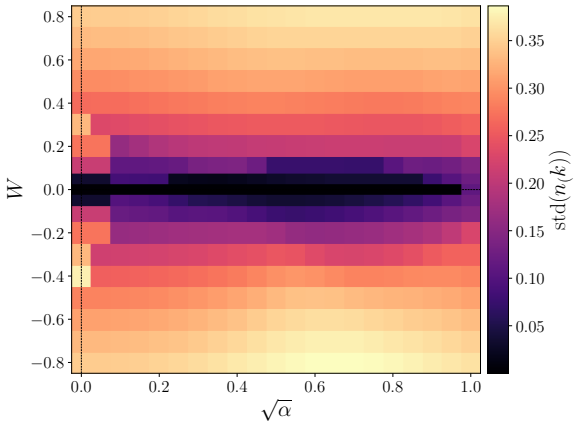


Figure 4.15: Standard deviation of $n(\mathbf{k})$ over the Brillouin zone as a function of α and W, at $\nu = \frac{1}{2}$.

Chapter 5

Results — $n = 1$ Landau level

Extending the model to the $n = 1$ Landau level modifies the interaction profile — it is effectively multiplied by the Laguerre polynomials. While the phase diagram is still expected to partition broadly into valley-polarized and unpolarized regimes, following a similar exchange-driven mechanism, the specific ground-state competition across (α, W) changes. Here, only the filling of $\nu = 3$ ($\nu_{n=1} = 1$) is studied as it is one where a qualitative difference compared to the lowest Landau level is expected — existence of a non-abelian FTI.

5.1 Anticipated phases

Across the parameter space, we anticipate the following phase competition:

- **Valley-polarized IQH state (low $|W|$, small α)**: At small inter-valley coupling and small bandwidth, similarly to the LLL case, complete valley polarization is suspected ($\nu_{n=1} = 1 + 0$). Analogously, integer filling in a single valley stabilizes an integer quantum Hall (IQH) state.
- **Paired non-abelian state ($\alpha = 1$, $W = 0$)**: In the decoupled limit each valley independently hosts a half-filled $n = 1$ Landau level ($\nu_{n=1} = \frac{1}{2} + \frac{1}{2}$). Instead of forming a composite Fermi liquid, the fermions become unstable to pairing, creating a non-abelian FTI consisting of two Moore–Read states, $|\mathrm{MR}\rangle \otimes \overline{|\mathrm{MR}\rangle}$ — found by Abouelkomsan and Fu [1] in a similar Landau level model. Since a single Moore–Read state has 2- and 6-fold degeneracy for odd and even number of fermions respectively, a ground-state degeneracy of an FTI formed from it will have a degeneracy of $2 \cdot 2 = 4$ or $6 \cdot 6 = 36$ respectively.
- **Depolarization to Fermi liquid (large W)**: As the bandwidth W increases, kinetic energy overcomes two-body interactions, eventually forcing a transition to a gapless, unpolarized Fermi liquid (FL).

5.1.1 Numerical criteria for defining phases

Similarly to the lowest Landau level, the phases are classified according to valley polarization, spectral gap, and the presence of spectral flow, using numeric criteria as in Table 5.1. Because of the thesis' focus on the non-abelian FTI (MR), additional analysis of the UP umbrella-region is not performed here.

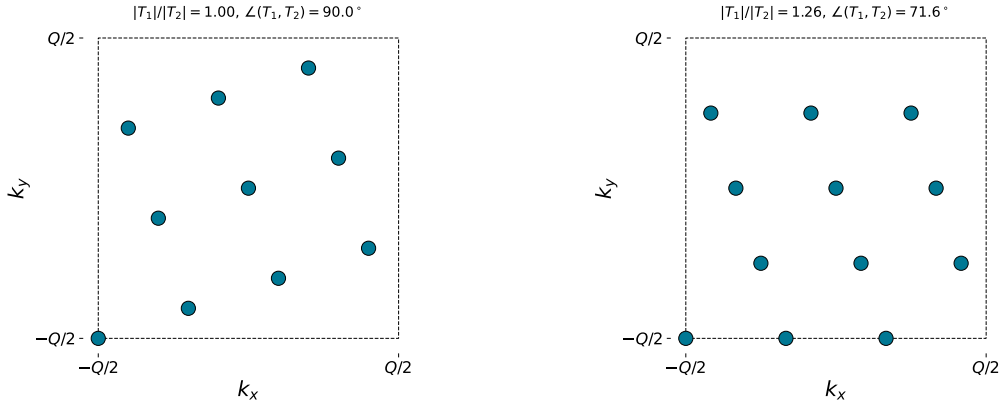
5.2 Results for $\nu_{n=1} = 1$ of 1LL ($\nu = 3$ overall)

Since the Moore–Read state is known to be influenced by the anisotropy of the real-space cluster on which it is defined, tilted boundary conditions were used to stabilize it. To detect

Label	P_z	Criterion
IQH	full	$\Delta_1 > 0$
UP	0	gapless (unresolved)
NTG	0	$\Delta_{36} > 10^{-3}$, no spectral flow
MR	0	$\Delta_{4/36} > 10^{-3}$, spectral flow

Table 5.1: Numerical criteria used for the primary phase classification of the $n = 1$ Landau level results at $\nu = 3$ (Fig. 5.2).

the characteristic dependence of the ground-state degeneracy on the fermion-number parity, systems with $N_\phi = 12$ and $N_\phi = 10$ per valley were studied ($N_e = 6$ and $N_e = 5$ per valley, respectively). Due to computational cost, finite size scaling was again not performed.



(a) $N_\phi = 10$ (b) $N_\phi = 12$

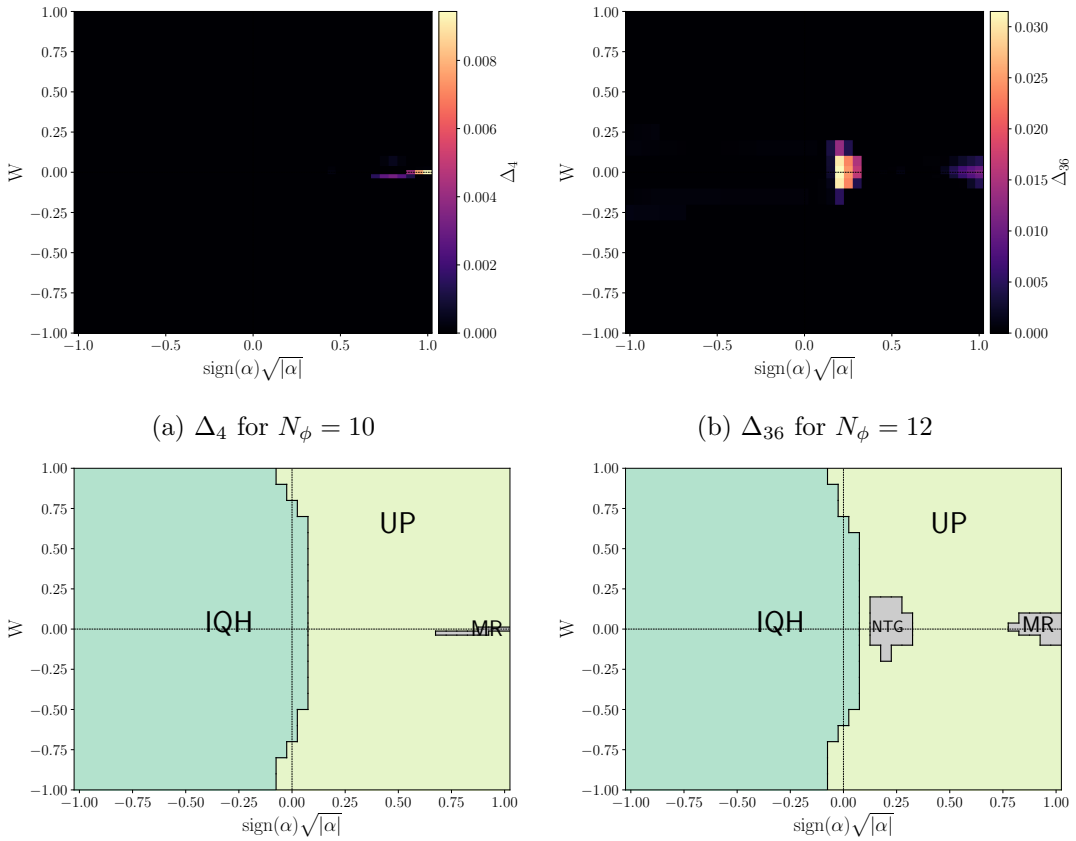
Figure 5.1: BZ sampling for the two system sizes used in the $n = 1$ LL calculations, and the properties of the corresponding real-space cluster vectors $\mathbf{T}_1$, $\mathbf{T}_2$.

The used tilted boundary condition BZ samplings are shown in Fig. 5.1.

Figure 5.2 shows the Δ_4 and Δ_{36} gaps for the two system sizes, as well as the schematic phase diagrams. The suspected results are confirmed: at low α and low W the system is in a $\Delta_1 > 0$ gapped, valley-polarized IQH phase, and the gap closes at large W as the system transitions to an unpolarized gapless Fermi liquid. At $\alpha = 1$, $W = 0$, the system is in a gapped, unpolarized phase with 4-fold and 36-fold degeneracy for $N_\phi = 10$ and $N_\phi = 12$ respectively, which is consistent with the expected non-abelian FTI phase. However, despite qualitative agreement of both system sizes, the stability of the MR phase against α and W differs significantly between them. This discrepancy is resolved through a supplementary calculation in a larger system with $N_\phi = 14$ in Appendix A, and due to this, the $N_\phi = 12$ results are deemed more reliable and used in following calculations. Interestingly, the stability against finite $|W|$ is very similar to the abelian FTI in the lowest Landau level. On the other hand, the stability against $\alpha \neq 1$ is slightly better, giving $\alpha_c \approx 0.64$, compared to $\alpha_c \approx 0.72$ for the FTI in the LLL examined in §4.3.

5.2.1 Spectral flow in the $N_\phi = 12$ system

To confirm the nature of the $\Delta_{36} > 0$ regions of the phase diagram, flux insertion was performed to check for spectral flow. Figure 5.3a shows flux-insertion for the $N_\phi = 12$ system at $\alpha = 1$, $W = 0.05$. The 36-fold degenerate ground-state manifold is clearly visible, and spectral flow is seen between groups of states in the manifold. However, the gap between the 36-fold degenerate



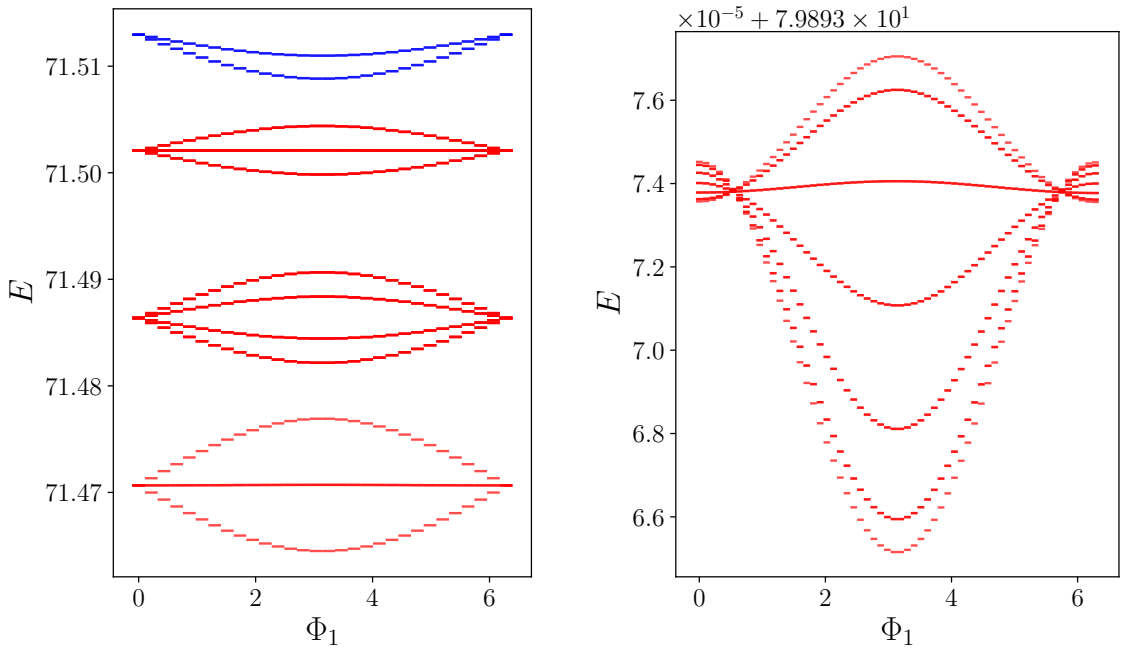
(a) Δ_4 for $N_\phi = 10$

(b) Δ_{36} for $N_\phi = 12$

(c) Phase diagram for $N_\phi = 10$

(d) Phase diagram for $N_\phi = 12$

Figure 5.2: Relevant spectral gaps and phase diagrams for the $n = 1$ Landau level at $\nu = 1$. UP: general unpolarized state. MR: non-abelian Moore–Read-like FTI. IQH: valley-polarized integer quantum Hall state. NTG: non-topological gapped state.



(a) $\alpha = 1.0$ (b) $\alpha = 0.07$

Figure 5.3: Phase winding for $N_\phi = 12$ at $W = 0.05$ in two Δ_{36} gapped phases.

ground-state manifold and the first excited state is in fact smaller than the spread of energies in the ground-state manifold itself. While the large spread is often interpreted as questionable stability of the state, here it might be caused by the Moore–Read state exhibiting significantly larger correlation length compared to the Laughlin state [17] — larger system sizes are needed to capture good degeneracy.

At the lower–α end of the unpolarized phase, the Δ_{36} gap is also significant (Fig. 5.2b). Under flux insertion (Fig. 5.3b), it is clear, however, that the spectral flow is not present in this region — the states don't permute under 2π flux insertion. Additionally the region is not adiabatically connected to the $\alpha = 1$, $W = 0$ point. The phase is not topological in nature, likely a charge density wave (CDW). It is not studied here further and is labeled as NTG on the phase diagram.

5.2.2 Stability of the LLL and $n = 1$ LL FTI phases — screening dependence

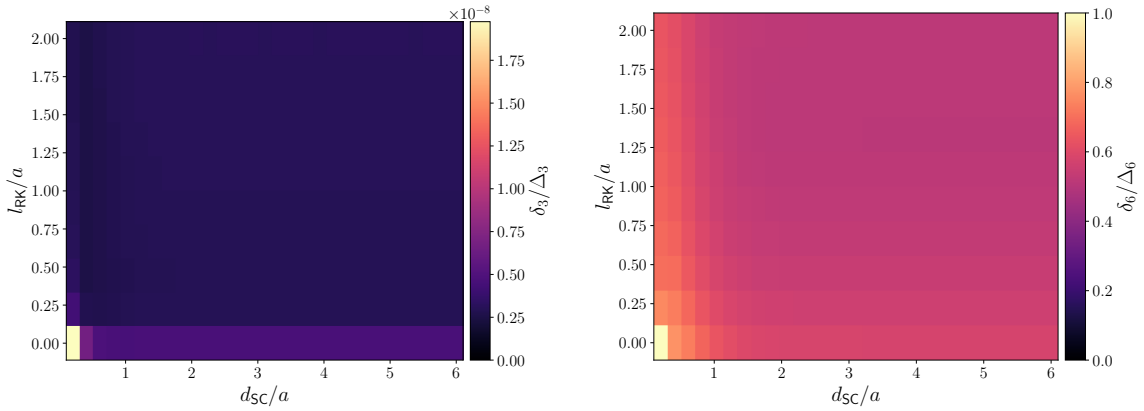
The ground-state degeneracy of the topologically nontrivial states becomes exact only in the thermodynamic limit. For finite systems, instead of the direct spectral gap, the ratio of the energy spread δ_N of the N-fold degenerate ground-state manifold to the Δ_N gap is often considered, with the premise that

$$\frac{\delta_N}{\Delta_N} = 0$$

indicates a perfectly degenerate ground state [24].

Since the non-abelian FTI phase has a large spread of energies in the ground-state manifold, its stability is tested against the screening parameters d_{SC} and l_{RK}. For simplicity, the decoupled limit $\alpha = 1$, $W = 0$ is considered. This means that each valley can be treated independently — the model is restricted to a single valley, where instead of 4-fold and 36-fold degeneracy, a 2-fold and 6-fold degeneracy is expected.

In Fig. 5.4b, the dependence of the δ_6/Δ_6 ratio on the screening parameters is shown. The ratio decreases monotonically with increasing d_{SC} and l_{RK}. For comparison, an analogous plot



(a) Lowest Landau level

(b) $n = 1$ Landau level

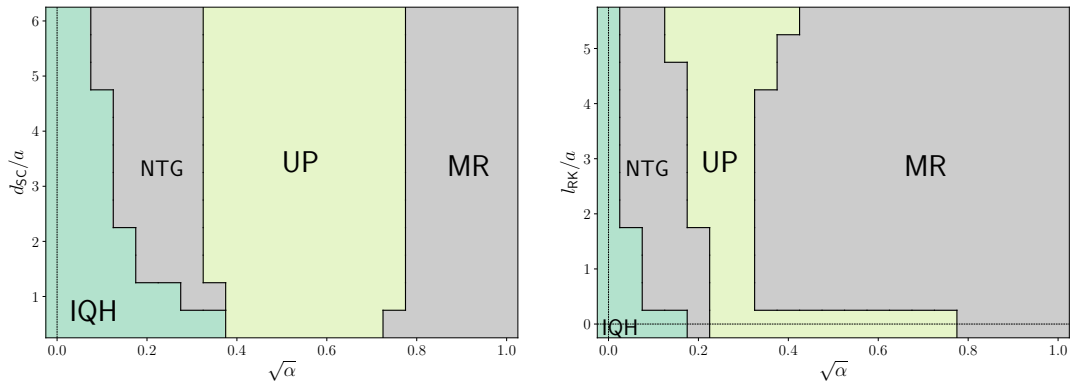
Figure 5.4: δ_n/Δ_n ratio in one decoupled $|\mathrm{FCI}\rangle$ state corresponding to $W = 0$ and $\alpha = 1$, as a function of d_{SC} and l_{RK} screening parameters.

for the δ_3/Δ_3 ratio in a single-valley $\nu = 1/3$ Laughlin-like FCI state in lowest Landau level is shown on Figure 5.4a. The ratio is much larger for the $\nu_{n=1} = 1/2$ Moore–Read state. While for the lowest Landau level FCI it is consistently close to 0, for the $n = 1$ Landau level it does not fall below the 0.5 threshold. The monotonicity suggests that for both the states in LLL and $n = 1$ LL the dependence of stability on screening is similar, while the difference in absolute values once again might indicate the difference in correlation lengths between Laughlin and Moore–Read states.

On the other hand, it is important to note that the stability of the FTI phase shown here at the $\alpha = 1$ point does not directly correspond to the extent of the FTI phase in the full (α, W) phase diagram. In Figure 4.9b a maximum of inter–valley coupling stability was shown at $l_{\mathrm{RK}} \neq 0$, while here, the δ/Δ ratio decreases monotonically with increasing l_{RK}. This further demonstrates the necessity of accounting for inter-valley interactions — a consideration addressed directly in the following section.

5.2.3 Screening of the interaction

The stability of the topological phases in $n = 1$ LL was tested against the two screening parameters.



(a) α vs. d_{SC}/a, at $l_{\mathrm{RK}} = 0$

(b) α vs. l_{RK}/a, at $d_{\mathrm{SC}} = 2a$

Figure 5.5: Phase diagrams of the $\nu = 3$ filling at $W = 0$ with respect to the screening parameters.

Figure 5.5a shows the extent of the phases at $W = 0$ as a function of α and the screening length d_{SC}. The conclusions are analogous to the LLL results. The α-stability of the IQH phase decreases monotonically with increasing d_{SC}, while no significant change in the extent of the MR (FTI) phase is seen.

Figure 5.5b shows the analogous dependence on l_{RK} at $W = 0$. Again, while the IQH phase becomes less stable with increasing l_{RK}, the non-abelian FTI phase instead shows a maximum of stability at non-zero l_{RK}, with the lowest transition found at $\alpha_c \approx 0.12$, which is much closer to the physically relevant point of $\alpha = 0$ than in the case of the abelian FTI in LLL.

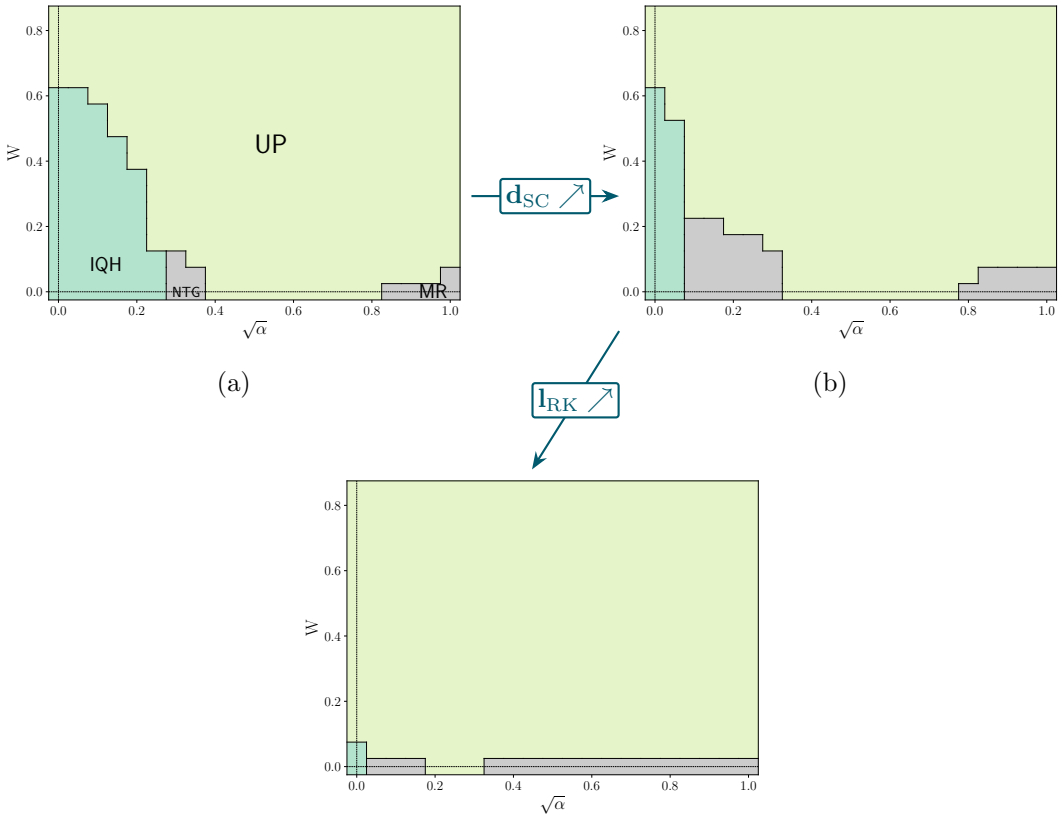


Figure 5.6: Phase diagrams of the $\nu = 3$ filling in the α–W plane at different d_{SC} and l_{RK}. Arrows indicate the direction of change of the parameters. (a) $d_{\mathrm{SC}} = 1a$, $l_{\mathrm{RK}} = 0$. (b) $d_{\mathrm{SC}} = 6a$, $l_{\mathrm{RK}} = 0$. (c) $d_{\mathrm{SC}} = 6a$, $l_{\mathrm{RK}} = 2a$.

Looking at full α–W phase diagrams at different screening parameters in Figure 5.6 one can again see that while the MR (FTI) responds only to l_{RK}, the W-extent of both topological phases significantly decreases at non-zero l_{RK}. The result is thus again qualitatively similar to the LLL screening analysis, however, with a comparatively larger α-extent in the non-abelian FTI examined here.

Chapter 6

Conclusions and outlook

6.1 Conclusions

This thesis studied a two-valley Landau-level model designed to capture the essential physics of correlated and topological phases in moiré transition metal dichalcogenides, using exact diagonalization on finite tori.

The resulting phase diagrams of studied systems are summarized in Figure 6.1. At filling $\nu = 2/3$, we found clear numerical evidence — via ground-state degeneracy and spectral flow — for a valley-polarized fractional quantum Hall state (FQH) at small inter-valley coupling α and small bandwidth W, and for a valley-unpolarized abelian fractional topological insulator (FTI) built from two time-reversed Laughlin copies at large α, consistent with theoretical expectations. Both phases were shown to survive a finite bandwidth (up to $|W| \approx 0.1$) and non-ideal inter-valley coupling (down to $\alpha \approx 0.72$ for the FTI), with the phase boundaries dependent on d_{SC} and l_{RK} screening parameters. In particular, non-zero l_{RK} shows improved stability against inter-valley interaction while exhibiting worse stability against finite bandwidth W. At $\nu = 1$, we recovered the expected valley-polarized integer quantum Hall state, together with signatures suggesting an IVC state at intermediate α, though we were not able to confirm this identification directly. Similarly, signatures suggesting composite Fermi liquids were found at $\nu = 1$ and $\nu = 1/2$. As expected, at all fillings the momentum occupation function suggests Fermi liquid behavior at large $|W|$. Additionally, all phase diagrams are approximately symmetric with respect to $W = 0$, consistent with the mapping between $W > 0$ and $W < 0$ Hamiltonians described in §2.3. For all the fillings phase diagrams were also sketched with modified interaction $U(\mathbf{q} = 0) = 0$. At all of the fillings this resulted in a valley polarized phase at large α up to $\alpha = 1$ and an unpolarized one at low α. In the polarized state, FQH and Fermi liquid were found at $\nu = 2/3$, IQH at $\nu = 1$ and CFL and FL at $\nu = 1/2$ — with transitions at similar $|W|$ as in the $U(\mathbf{q} = 0) \neq 0$ case. The unpolarized state was not studied in detail.

An analytic Hartree–Fock treatment of the valley-depolarization transition, developed for both the α-driven and W-driven cases, reproduced the correct qualitative trends of the numerical phase diagrams — including the direction of the phase boundaries with d_{SC} and l_{RK} — while systematically overestimating the critical α_c and W_c by up to an order of magnitude, largely due to the approximation neglecting quantum fluctuations. However, the predictions are improved compared to those of Raines, Glazman and Chubkov [39] — due to the inclusion of cosine dispersion specific for our model.

Calculations in $n = 1$ Landau level were performed at $\nu = 3$ filling, with tilted boundary conditions to alleviate the issues caused by cluster anisotropy. As suspected, the integer quantum Hall state was found in the valley polarized region at low α. Near the $\alpha = 1$ limit a non-abelian FTI composed of time-reversed copies of Moore–Read states is found. It has similar stability against non-zero W to the Laughlin-like FTI in the lowest Landau level; however its stability against non-zero inter-valley coupling is improved (down to $\alpha_c \approx 0.64$, and $\alpha_c \approx 0.12$

at finite l_{RK}). For both the abelian and non-abelian FTI states, the Rytova–Keldysh screening length l_{RK} is found to have a non-monotonic effect on stability: a maximum of α-stability is reached at finite l_{RK}, at the cost of reduced stability against bandwidth W.

Taken together, these results support the picture of moiré TMDs as a magnetic-field-free platform for realizing both fractional Chern insulator and fractional topological insulator physics, including non-abelian variants, and provide a systematic account of how this physics changes with bandwidth, inter-valley coupling, and dielectric screening within a single, tunable model.

6.2 Outlook

The main open questions raised by this work are left for future work.

- **Finite-size scaling.** All results in this thesis are obtained by exact diagonalization on a single system size per filling ($N_\phi = 12$ for the lowest Landau level, with a separate tilted geometry for the momentum-occupation calculations, and $N_\phi = 10$ and $N_\phi = 12$ for the $n = 1$ Landau level). A rigorous identification of topological order requires finite-size scaling — confirming that the many-body gap remains open and the ground-state degeneracy persists as the system size increases — which was not performed here due to computational cost. Performing this scaling, particularly for the α- and W-driven depolarization transitions at $\nu = 2/3$ and $\nu = 1$, would let the phase boundaries reported here (e.g. α_c, W_c) be extrapolated to the thermodynamic limit.

- **Momentum grid choice.** The untilted rectangular momentum grid used by default in Chapter 4 is known to bias the competition between topological and charge-density-wave-like states relative to more isotropic tilted grids [26]. This choice was made for simplicity and consistency across the phase diagrams, so the reported phase boundaries may not be fully representative of the thermodynamic-limit behavior. Repeating the $\nu = 2/3$ and $\nu = 1$ phase diagrams with tilted boundary conditions throughout, as was done for the momentum-occupation calculations and for Chapter 5, would clarify which features are physical and which are artifacts of the cluster shape, as was done explicitly for the $\nu = 1$ case in §4.4.1.

- **Treatment of the $q = 0$ interaction component.** Two variants of the interaction were considered in this thesis, differing in their treatment of the $U(\mathbf{q} = 0)$ term. The qualitative difference between the two resulting phase diagrams throughout Chapter 4 suggests this choice is not negligible, and a more careful treatment of the neutralizing background is left for future work.

- **Model scope.** The model of Chapter 2 captures the two-valley Landau-level physics of moiré TMDs at a single magic angle, projected onto a single Landau-level-like band per valley, and neglects Landau-level mixing. Real moiré TMD flat bands deviate from ideal Landau levels in ways not captured by this construction, such as non-uniform Berry curvature, deviations of the form factor from the ideal Gaussian form, and Landau level mixing. Another simplification is the use of a square lattice and a square Brillouin zone instead of a hexagonal one usually found in moiré materials. Quantitative comparison to specific materials and twist angles would require using actual continuum-model form factors rather than the idealized Landau-level ones used here.

- **Unresolved intermediate phases.** Several regions of the phase diagrams — the candidate IVC regions inferred from flat momentum occupation at $\nu = 2/3$ and $\nu = 1$ (§4.4.1, §4.3.3), and the $|\mathrm{CFL}\rangle \otimes \overline{|\mathrm{CFL}\rangle}$–$|\mathrm{FL}\rangle \otimes \overline{|\mathrm{FL}\rangle}$ transition at $\nu = 1$ (§4.4.1) — were identified only indirectly, through the degeneracy/spectral-flow/occupation diagnostics of §3.4.

Some flat-$n(\mathbf{k})$ regions in the summary phase diagram (Fig. 6.1) remain unidentified even indirectly. A more conclusive identification of these states, and of the non-abelian FTI of Chapter 5, whose ground-state energy spread is comparable to its many-body gap even at the largest system size studied, would require computing their overlap with candidate trial wavefunctions.

- **Analytic treatment of transitions out of topological phases.** The mean-field treatment of §4.2 was developed specifically for the valley-depolarization transitions. The other transitions found in this thesis — the W-driven transitions out of the FQH and FTI states at $\nu = 2/3$ and out of the non-abelian FTI at $n = 1$, as well as the α-driven transition out of the FTI state — were only detected numerically, through the closing of the many-body gap and the loss of spectral flow, without any corresponding analytic treatment of their location or nature. A corresponding analytic treatment of these transitions was not attempted and is left for future work.

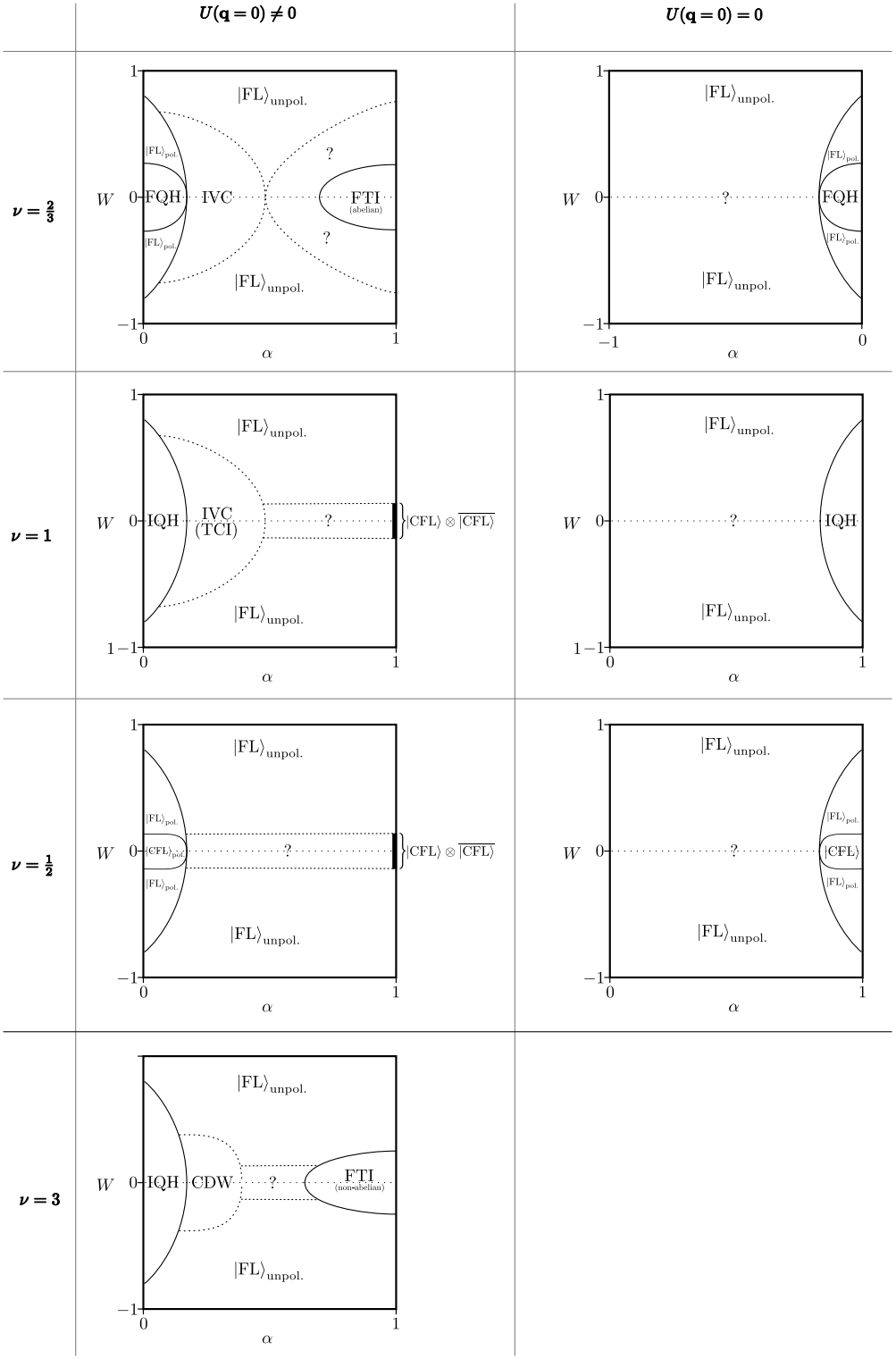


Figure 6.1: Summary of phase diagrams obtained throughout the thesis for different filling fractions and interaction setups. Solid lines represent directly confirmed phases, whereas dashed lines, phases inferred from indirect signatures, known limits and literature. Question marks denote regions with flat momentum occupation that were not identified.

Bibliography

[1] Ahmed Abouelkomsan and Liang Fu. "Non-abelian spin Hall insulator". In: *Physical Review Research* 7.2 (2025), p. 023083. DOI: 10.1103/PhysRevResearch.7.023083.

[2] Maissam Barkeshli and John McGreevy. "Continuous transitions between composite Fermi liquid and Landau Fermi liquid: A route to fractionalized Mott insulators". In: *Physical Review B—Condensed Matter and Materials Physics* 86.7 (2012), p. 075136. DOI: 10.1103/PhysRevB.86.075136.

[3] B Andrei Bernevig and N Regnault. "Emergent many-body translational symmetries of Abelian and non-Abelian fractionally filled topological insulators". In: *Physical Review B—Condensed Matter and Materials Physics* 85.7 (2012), p. 075128. DOI: 10.1103/PhysRevB.85.075128.

[4] Nick Bultinck, Shubhayu Chatterjee, and Michael P. Zaletel. "Mechanism for Anomalous Hall Ferromagnetism in Twisted Bilayer Graphene". In: *Phys. Rev. Lett.* 124 (16 2020), p. 166601. DOI: 10.1103/PhysRevLett.124.166601.

[5] Jiaqi Cai et al. "Signatures of fractional quantum anomalous Hall states in twisted MoTe2". In: *Nature* 622.7981 (2023), pp. 63–68. DOI: 10.1038/s41586-023-06289-w.

[6] Yuan Cao et al. "Correlated insulator behaviour at half-filling in magic-angle graphene superlattices". In: *Nature* 556.7699 (2018), pp. 80–84. DOI: 10.1038/nature26154.

[7] Yuan Cao et al. "Unconventional superconductivity in magic-angle graphene superlattices". In: *Nature* 556.7699 (2018), pp. 43–50. DOI: 10.1038/nature26160.

[8] Hua Chen and Kun Yang. "Interaction-driven quantum phase transitions in fractional topological insulators". In: *Physical Review B—Condensed Matter and Materials Physics* 85.19 (2012), p. 195113. DOI: 10.1103/PhysRevB.85.195113.

[9] R Coehoorn, C Haas, and RA De Groot. "Electronic structure of MoSe 2, MoS 2, and WSe 2. II. The nature of the optical band gaps". In: *Physical Review B* 35.12 (1987), p. 6203. DOI: 10.1103/PhysRevB.35.6203.

[10] Valentin Crépel and Liang Fu. "Anomalous Hall metal and fractional Chern insulator in twisted transition metal dichalcogenides". In: *Physical Review B* 107.20 (2023), p. L201109. DOI: 10.1103/PhysRevB.107.L201109.

[11] Pierluigi Cudazzo, Ilya V Tokatly, and Angel Rubio. "Dielectric screening in two–dimensional insulators: Implications for excitonic and impurity states in graphane". In: *Physical Review B—Condensed Matter and Materials Physics* 84.8 (2011), p. 085406. DOI: 10.1103/PhysRevB.84.085406.

[12] Sahana Das, Glenn Wagner, and Titus Neupert. "Fractional topological insulators at odd-integer filling: Phase diagram of two-valley quantum Hall model". In: *arXiv preprint arXiv:2509.16335* (2025). DOI: 10.48550/arXiv.2509.16335.

[13] Sankar Das Sarma, Michael Freedman, and Chetan Nayak. "Topologically protected qubits from a possible non-Abelian fractional quantum Hall state". In: *Physical review letters* 94.16 (2005), p. 166802. DOI: 10.1103/PhysRevLett.94.166802.

[14] Trithep Devakul et al. "Magic in twisted transition metal dichalcogenide bilayers". In: *Nature communications* 12.1 (2021), p. 6730. DOI: 10.1038/s41467-021-27042-9.

[15] Junkai Dong et al. "Composite Fermi liquid at zero magnetic field in twisted MoTe 2". In: *Physical Review Letters* 131.13 (2023), p. 136502. DOI: 10.1103/PhysRevLett.131.136502.

[16] Federico Escudero et al. "Designing moiré patterns by strain". In: *Physical Review Research* 6.2 (2024), p. 023203. DOI: 10.1103/PhysRevResearch.6.023203.

[17] B Estienne, N Regnault, and B Andrei Bernevig. "Correlation lengths and topological entanglement entropies of unitary and nonunitary fractional quantum hall wave functions". In: *Physical review letters* 114.18 (2015), p. 186801. DOI: 10.1103/PhysRevLett.114.186801.

[18] F. D. M. Haldane. "Model for a Quantum Hall Effect without Landau Levels: Condensed-Matter Realization of the "Parity Anomaly"". In: *Phys. Rev. Lett.* 61 (18 1988), pp. 2015–2018. DOI: 10.1103/PhysRevLett.61.2015.

[19] J. K. Jain. "Composite-fermion approach for the fractional quantum Hall effect". In: *Phys. Rev. Lett.* 63 (2 1989), pp. 199–202. DOI: 10.1103/PhysRevLett.63.199.

[20] C. L. Kane and E. J. Mele. "Quantum Spin Hall Effect in Graphene". In: *Phys. Rev. Lett.* 95 (22 2005), p. 226801. DOI: 10.1103/PhysRevLett.95.226801.

[21] Leonid Veniaminovich Keldysh. "Coulomb interaction in thin semiconductor and semimetal films". In: *Selected Papers of Leonid V Keldysh*. World Scientific, 2024, pp. 155–158. DOI: 10.1142/9789811279461_0024.

[22] Andor Kormányos et al. "k· p theory for two-dimensional transition metal dichalcogenide semiconductors". In: *2D Materials* 2.2 (2015), p. 022001. DOI: 10.1088/2053-1583/2/2/022001.

[23] Yves H Kwan et al. "Textured exciton insulators". In: *Physical Review B* 112.3 (2025), p. 035129. DOI: 10.1103/4k6g-hy6y.

[24] Yves H Kwan et al. "When Could Abelian Fractional Topological Insulators Exist in Twisted MoTe _2(and Other Systems)". In: *arXiv preprint arXiv:2407.02560* (2024). DOI: 10.48550/arXiv.2407.02560.

[25] Cornelius Lanczos. "An iteration method for the solution of the eigenvalue problem of linear differential and integral operators". In: *Journal of Research of the National Bureau of Standards* 45.4 (1950), pp. 255–282. DOI: 10.6028/jres.045.026.

[26] Andreas M Läuchli et al. "Hierarchy of fractional Chern insulators and competing compressible states". In: *Physical Review Letters* 111.12 (2013), p. 126802. DOI: 10.1103/PhysRevLett.111.126802.

[27] Robert B Laughlin. "Anomalous quantum Hall effect: An incompressible quantum fluid with fractionally charged excitations". In: *Physical Review Letters* 50.18 (1983), p. 1395. DOI: 10.1103/PhysRevLett.50.1395.

[28] Bohao Li and Fengcheng Wu. "Variational mapping of Chern bands to Landau levels: Application to fractional Chern insulators in twisted MoTe 2". In: *Physical Review B* 111.12 (2025), p. 125122. DOI: 10.1103/PhysRevB.111.125122.

[29] Qiuyang Li et al. “Twist engineering of anisotropic excitonic and optical properties of a two-dimensional magnetic semiconductor”. In: *Physical Review Letters* 135.15 (2025), p. 156901. DOI: 10.1103/zbbr-5mgv.

[30] Kin Fai Mak and Jie Shan. “Semiconductor moiré materials”. In: *Nature Nanotechnology* 17.7 (2022), pp. 686–695. DOI: 10.1038/s41565-022-01165-6.

[31] IA McDonald and FDM Haldane. “Topological phase transition in the $\nu = 2/3$ quantum Hall effect”. In: *Physical Review B* 53.23 (1996), p. 15845. DOI: 10.1103/PhysRevB.53.15845.

[32] Gregory Moore and Nicholas Read. “Nonabelions in the fractional quantum hall effect”. In: *Nuclear Physics B* 360.2 (1991), pp. 362–396. DOI: https://doi.org/10.1016/0550-3213(91)90407-O.

[33] Chetan Nayak et al. “Non-Abelian anyons and topological quantum computation”. In: *Reviews of Modern Physics* 80.3 (2008), pp. 1083–1159. DOI: 10.1103/RevModPhys.80.1083.

[34] Titus Neupert et al. “Fractional quantum Hall states at zero magnetic field”. In: *Physical review letters* 106.23 (2011), p. 236804. DOI: 10.1103/PhysRevLett.106.236804.

[35] Kostya S Novoselov et al. “Two-dimensional gas of massless Dirac fermions in graphene”. In: *nature* 438.7065 (2005), pp. 197–200. DOI: 10.1038/nature04233.

[36] Masaki Oshikawa and T Senthil. “Fractionalization, topological order, and quasiparticle statistics”. In: *Physical review letters* 96.6 (2006), p. 060601. DOI: 10.1103/PhysRevLett.96.060601.

[37] Haining Pan, Fengcheng Wu, and Sankar Das Sarma. “Band topology, Hubbard model, Heisenberg model, and Dzyaloshinskii-Moriya interaction in twisted bilayer WSe 2”. In: *Physical Review Research* 2.3 (2020), p. 033087. DOI: 10.1103/PhysRevResearch.2.033087.

[38] Heonjoon Park et al. “Observation of fractionally quantized anomalous Hall effect”. In: *Nature* 622.7981 (2023), pp. 74–79. DOI: 10.1038/s41586-023-06536-0.

[39] Zachary M Raines, Leonid I Glazman, and Andrey V Chubukov. “Unconventional discontinuous transitions in a two-dimensional system with spin and valley degrees of freedom”. In: *Physical Review B* 110.15 (2024), p. 155402. DOI: 10.1103/PhysRevB.110.155402.

[40] Nicholas Read and Dmitry Green. “Paired states of fermions in two dimensions with breaking of parity and time-reversal symmetries and the fractional quantum Hall effect”. In: *Physical Review B* 61.15 (2000), p. 10267. DOI: 10.1103/PhysRevB.61.10267.

[41] Nicolas Regnault et al. *DiagHam: A suite of C++ codes for exact diagonalization of quantum many-body systems*. Online software repository. 2024. URL: https://www.nick-ux.org/diagham/index.php/Main_Page.

[42] Nicolas Regnault and B Andrei Bernevig. “Fractional Chern insulator”. In: *Physical Review X* 1.2 (2011), p. 021014. DOI: 10.1103/PhysRevX.1.021014.

[43] Cecile Repellin, B Andrei Bernevig, and Nicolas Regnault. “Z 2 fractional topological insulators in two dimensions”. In: *Physical Review B* 90.24 (2014), p. 245401. DOI: 10.1103/PhysRevB.90.245401.

[44] Edward H Rezayi and F Duncan M Haldane. “Incompressible paired Hall state, stripe order, and the composite fermion liquid phase in half-filled Landau levels”. In: *Physical review letters* 84.20 (2000), p. 4685. DOI: 10.1103/rezayi2000incompressible.

[45] Natalia S Rytova. "Screened potential of a point charge in a thin film". In: *Moscow University Physics Bulletin* 22.3 (1967), p. 18. DOI: 10.48550/arXiv.1806.00976.

[46] D. N. Sheng et al. "Fractional quantum Hall effect in the absence of Landau levels". In: *Nature Communications* 2.1 (2011), p. 389. DOI: 10.1038/ncomms1380.

[47] Horst Ludwig Stormer. "Two-dimensional electron correlation in high magnetic fields". In: *Physica B: Condensed Matter* 177.1-4 (1992), pp. 401–408. DOI: 10.1016/0921-4526(92)90138-I.

[48] M Storni, RH Morf, and S Das Sarma. "Fractional quantum hall state at ν= 5 2 and the moore-read pfaffian". In: *Physical review letters* 104.7 (2010), p. 076803. DOI: 10.1103/PhysRevLett.104.076803.

[49] David Tong. "Lectures on the quantum Hall effect". In: *arXiv preprint arXiv:1606.06687* (2016). DOI: 10.48550/arXiv.1606.06687.

[50] Klaus Von Klitzing. "The quantized Hall effect". In: *Reviews of Modern Physics* 58.3 (1986), p. 519. DOI: 10.1103/RevModPhys.58.519.

[51] Yiping Wang et al. "Candidate for a Fractional Topological Insulator in Twisted $MoTe_2$". In: *Phys. Rev. X* 16 (3 2026), p. 031009. DOI: 10.1103/bvrb-z4hj.

[52] R. Willett et al. "Observation of an even-denominator quantum number in the fractional quantum Hall effect". In: *Phys. Rev. Lett.* 59 (15 1987), pp. 1776–1779. DOI: 10.1103/PhysRevLett.59.1776.

[53] Fengcheng Wu et al. "Hubbard model physics in transition metal dichalcogenide moiré bands". In: *Physical review letters* 121.2 (2018), p. 026402. DOI: 10.1103/PhysRevLett.121.026402.

[54] Di Xiao et al. "Coupled spin and valley physics in monolayers of MoS 2 and other group-VI dichalcogenides". In: *Physical review letters* 108.19 (2012), p. 196802. DOI: 10.1103/PhysRevLett.108.196802.

[55] Cheng Xu et al. "Multiple Chern bands in twisted MoTe 2 and possible non-Abelian states". In: *Physical Review Letters* 134.6 (2025), p. 066601. DOI: 10.1103/PhysRevLett.134.066601.

[56] Fan Xu et al. "Observation of integer and fractional quantum anomalous Hall effects in twisted bilayer MoTe 2". In: *Physical Review X* 13.3 (2023), p. 031037. DOI: 10.1103/PhysRevX.13.031037.

[57] Yihang Zeng et al. "Thermodynamic evidence of fractional Chern insulator in moiré MoTe2". In: *Nature* 622.7981 (2023), pp. 69–73. DOI: 10.1038/s41586-023-06452-3.

[58] Siwen Zhao et al. "Probing the fractional quantum Hall phases in valley-layer locked bilayer MoS _{2}". In: *arXiv preprint arXiv:2308.02821* (2023). DOI: 10.48550/arXiv.2308.02821.

[59] Bo Zou, Anzhuoer Li, and A. H. MacDonald. "Valley order in moiré topological insulators". In: *Phys. Rev. B* 113 (24 2026), p. 245141. DOI: 10.1103/kl58-rj3r.

Appendix A

Reconciliation of $\nu = 3$ phase diagram results

Section 5.2 showed a significant discrepancy between the stability of the Moore–Read-like FTI obtained from the $N_\phi = 10$ and $N_\phi = 12$ systems. Here, an additional calculation is done on a tilted cluster with $N_\phi = 14$ and 7 electrons per valley, in order to elucidate this contradictory result. Due to the significant computational cost, only the $W \in [0.0, 0.2]$, $\alpha \in [0.0, 1.0]$ part of the phase diagram is calculated. The parameters of the Hamiltonian are the same as those used in §5.2. As in the $N_\phi = 10$ case, the number of electrons per valley is odd, meaning the candidate Moore–Read-like FTI is again expected to have a $2 \times 2 = 4$-fold ground-state degeneracy.

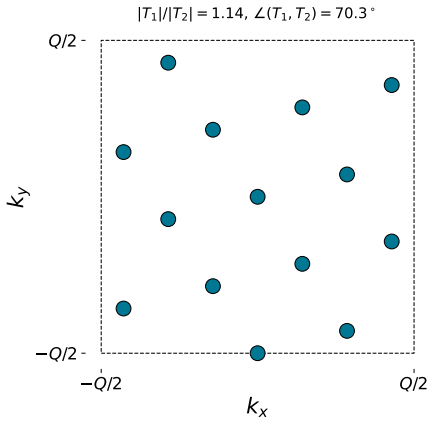
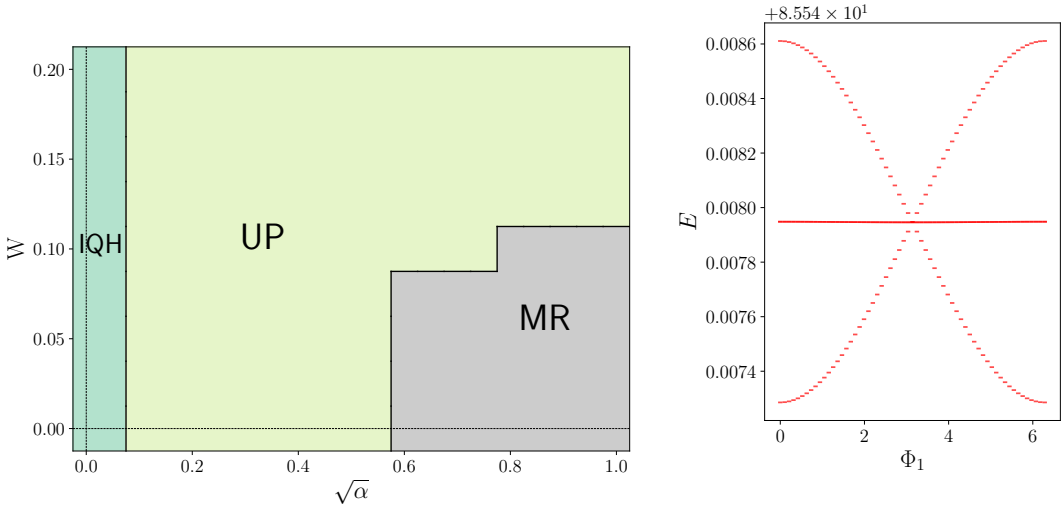


Figure A.1: Sampling of the Brillouin zone used for the supplementary odd-sector Moore–Read FTI calculation with $N_\phi = 14$, and properties of the corresponding $\mathbf{T}_i$ real-space vectors.

Figure A.1 shows the tilted-boundary-condition sampling of the Brillouin zone used for the $N_\phi = 14$ calculation. Compared to $N_\phi = 10$ in Figure 5.1a, the $N_\phi = 14$ cluster has a larger $|\mathbf{T}_1|/|\mathbf{T}_2|$ ratio, however the angle between the vectors is smaller, meaning the sampling is closer to the more isotropic hexagonal-lattice case. It cannot be deemed better than the $N_\phi = 10$ case on these grounds alone. It is nonetheless understood that the odd- and even-fermion-number phase diagrams should converge in the thermodynamic limit. Taking into account the 40% increase in system size between the two clusters, together with the expected long correlation length of the Moore–Read state, the $N_\phi = 14$ results can *a priori* be expected to be more accurate and trustworthy than those at $N_\phi = 10$.

Figure A.2a shows the resulting phase diagram of the $N_\phi = 14$ system. The phase diagram looks qualitatively similar to that at $N_\phi = 10$. The main difference is the extent of the MR



(a) Phase diagram

(b) Flux insertion at $W = 0.05$, $\alpha = 1.0$

Figure A.2: Phase diagram of the $N_\phi = 14$, $\nu = 3$ system and flux-insertion plot of the FTI at $\alpha = 1$.

(Moore–Read-like FTI) phase, which now shows notably improved stability against nonzero bandwidth W. Furthermore, the extent of the phase agrees well with the phase diagram of the same filling at $N_\phi = 12$ in Figure 5.2d. In light of this agreement between the two larger system sizes, the $N_\phi = 12$ results are treated as the more reliable estimate of the phase boundary and are used for the remainder of the analysis of the MR state in Chapter 5.